Information Processing and Management 61 (2024) 103790

Contents lists available at ScienceDirect

Information Processing and Management

journal homepage: www.elsevier.com/locate/ipm

Simplices-based higher-order enhancement graph neural network for multi-behavior recommendation

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ARTICLE INFO

Keywords:
Multi-behavior recommendations
Higher-order enhancement
Graph neural network
Implicit relationships

ABSTRACT

Multi-behavior recommendations effectively integrate various types of behaviors and have been proven to enhance recommendation performance. However, existing researches primarily focus on distinguishing between various behaviors, neglecting the exploration of common representations within each behavior that might reflect individual preferences from different perspectives. Meanwhile, interactions within each behavior remain sparse; how to learn effective information from limited data poses a significant challenge. In this study, we propose a simplices-based higher-order enhancement graph neural network for multi-behavior recommendations, HEM-GNN. Specifically, we adopt a supervised method to distinguish the importance of different behaviors and perform inter-behavior representation learning. Meanwhile, for each behavior, we define implicit relationships to mitigate data sparsity, and then aggregate information from nodes within simplices to extract their higher-order commonalities. Finally, HEM-GNN leverages these representations to make recommendations. Through experiments on three public datasets (Taobao, Beibei, and IJCAI), HEM-GNN demonstrates better performance compared to 10 baseline algorithms. It outperforms state-of-the-art models by margins ranging from 8.99% to 10.58% in HR@K and 8.18% to 9.69% in NDCG@K, highlighting the significance of higher-order features in multi-behavior recommendations. The model and datasets are released at: https://github.com/SamuelZack/MultiRec.

1. Introduction

As an effective way to alleviate information overload, personalized recommender systems based on collaborative filtering (CF) have been widely applied across various domains (Dai, Zhang, Pan, & Ming, 2019; Qiao, Zhou, Luo, & Wen, 2023; Xia, Huang, Xu, Zhao, et al., 2022). CF-based recommendation models leverage users' historical behaviors, such as purchase records, to mine their interests and estimate their preferences. However, the recommendation performance of these models is severely constrained by issues such as sparse data and cold start. For instance, on an online shopping website, it is difficult to accurately recommend products to new users without any historical purchase records, as illustrated in Fig. 1(a). Multi-behavior recommendation methods can gather and utilize users' auxiliary behaviors, such as browsing and bookmarking, to predict their preferences and provide accurate recommendations for the target behavior, *e.g.*, purchasing, as demonstrated in Fig. 1(b). Research has shown that auxiliary behaviors not only partially reflect users' preferences, but also tend to be more diverse than the target behavior (Jin, Gao, He, Jin, & Li, 2020).

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https://doi.org/10.1016/j.ipm.2024.103790
Received 9 December 2023; Received in revised form 30 April 2024; Accepted 19 May 2024
Available online 29 May 2024
0306-4573/

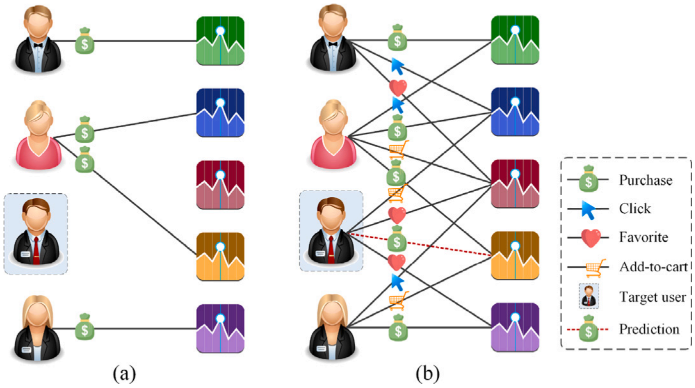


Fig. 1. Diagram of single- and multi-behavior interactions.

In the multi-behavior recommendation field, the most straightforward approach is to treat all types of behaviors uniformly, which can greatly enrich sparse data (Li, Ma, Zhang, Jin, & Li, 2023; Singh & Gordon, 2008). However, this approach overlooks the differences between various behaviors, such as purchasing and browsing, resulting in inconsistencies with real-world scenarios. As a consequence, effectively exploring user preferences under multiple behaviors becomes challenging, which in turn limits the recommendation performance. To distinguish the semantics of different behaviors, most multi-behavior recommendation models adopt an approach where each behavior is independently modeled and then integrated (Xia et al., 2020; Xu, Wang, Wu, et al., 2023).

However, existing multi-behavior recommendation models face the following limitation: they primarily focus on distinguishing among different behaviors but fail to accurately capture individual representations within behaviors. It is well known that accurately learning individual features within behaviors is crucial for effectively fusing multiple behaviors. Nonetheless, current approaches typically learn intra-behavior representations by broadly aggregating the neighbor information of users/items. For a given behavior, interactions are often sparse, and crudely aggregating neighbors with lower-order relationships (pairwise interactions) may introduce noise and lead to feature smoothing, thereby limiting the performance of multi-behavior recommendation models.

To address the above issues, we aim to explore higher-order relationships (group interactions) of individuals to enhance their common features within behaviors, and then perform multi-behavior fusion to improve recommendation accuracy. Therefore, we propose a novel multi-behavior recommendation model called the simplices-based higher-order enhancement graph neural network (HEM-GNN). HEM-GNN adopts a dual-tower architecture to independently capture the differences among various behaviors and the commonalities of individuals within each behavior. (1) To uncover the disparities among behaviors, we utilize a supervised approach for determining the significance (weight) of different behaviors. Subsequently, by leveraging these weights, we perform embedding learning within a multi-behavior context, thereby revealing the distinctions between auxiliary behaviors and the target behavior. (2) To learn the commonalities within behaviors, we define a computational method for implicit relationships to enrich sparse behavioral data. Subsequently, we aggregate nodes within simplices to obtain higher-order common representations. HEM-GNN utilizes these learned representations to achieve accurate recommendations. To summarize, the main contributions of this work are as follows:

- In this paper, we propose a simplices-based higher-order enhancement graph neural network, called HEM-GNN, aimed at enhancing the accuracy of multi-behavior recommendations. HEM-GNN can capture the differences between different behaviors and the higher-order commonalities of individuals within each behavior. This study reveals the importance of higher-order features in improving recommendation accuracy.
- To alleviate the data sparsity issue within behaviors, we propose an iterative calculation method for implicit relationships. These implicit relationships can not only effectively complement sparse explicit relationships, but also filter out noise, which has a positive effect on extracting higher-order behavioral features.
- To accurately explore higher-order commonalities of users, we design a higher-order relationship aggregation method. After enriching and filtering sparse data, we identify simplices in the behavioral graph and aggregate nodes within them to generate higher-order features. This method allows us to learn high-quality higher-order features from sparse data.
- We conduct a series of experiments to evaluate the performance of HEM-GNN on three real-world datasets. The results demonstrate that compared with other methods, HEM-GNN achieves superior recommendation performance. This performance advantage also extends to data sparse and cold start scenarios.

The remaining structure of this paper is as follows. Section 2 presents a concise overview of the related work. In Section 3, We offer the research objectives of the model. Section 4 describes the methodology and a detailed explanation of HEM-GNN. In Section 5, the experimental results are provided. Section 6 displays the discussion and implications of our study. Finally, in Section 7, we conclude the paper and discuss future outlooks.

2. Related work

2.1. Multi-behavior recommendation

Multi-behavior recommendation is a new branch in the field of recommender systems. Compared to single-behavior systems (Ding et al., 2021; Yu et al., 2024), incorporating additional behaviors effectively mitigates the problem of data sparsity and aids in predicting target behavior (Xin et al., 2023). Therefore, it can greatly improve recommendation performance (Chen, Zhang, Zhang, Liu, & Ma, 2020; Wang, Liu, Han, & Shi, 2021). According to existing research, we can classify multi-behavior recommendation models into two primary categories: shallow and deep multi-behavior recommendations.

Shallow multi-behavior recommendation models rely on traditional machine learning techniques, *i.e.*, shallow methods, to extract and exploit various types of behaviors. In particular, Zhao, Cheng, Hong, and Chi (2015) extended matrix factorization to multi-behavior recommendations by using shared embeddings and proposed the Collective Matrix Factorization (CMF) model. Moreover, several studies have explored different sampling strategies for extracting multi-behavioral information (Yan, Cheng, et al., 2023). For example, Loni, Pagano, Larson, and Hanjalic (2016) devised a negative sampling strategy tailored for various types of interactions, thus effectively extending the Bayesian Personalized Ranking (BPR) model (Rendle, Freudenthaler, Gantner, & Schmidt-Thieme, 2009) into the realm of multi-behavior recommendations. Expanding on this work, Ding et al. (2018) and Qiu et al. (2018) developed a negative sampler and an adaptive sampling strategy, respectively, further enhancing recommendation performance. Constrained by shallow techniques, these models struggle to capture deeper dependencies among various types of behaviors, leading to suboptimal performance.

Deep multi-behavior recommendation models employ neural networks and other deep learning architectures to capture intricate correlations within multi-behavioral data. Compared with shallow models, they generally exhibit superior performance. For instance, Xia et al. (2020) developed a Memory-Augmented Transformer Network (MATN), which explicitly encodes multiple behavioral relationships by retaining cross-type collaborative signals and contextual information. Meanwhile, both Zhang, Mao, Cao, and Xu (2020) and Xia, Huang, Xu, Dai, and Bo (2022) proposed multi-behavior recommendation models based on graph neural networks. These models fully consider different interaction patterns and the interdependence of potential cross-type behaviors, highlighting the significance of various behaviors in predicting user preferences. Further, Meng et al. (2023) introduced a Projection-based Transfer Network (PTN) for multi-behavior recommendations, which utilizes a projection mechanism to model the correlation between upstream and downstream behaviors. Similar to Meng et al. (2023), Cheng et al. (2023) addressed the feature transmission issue between upstream and downstream behaviors using a cascading graph convolutional network, thus presenting a multi-behavior recommendation model. Inspired by Cheng et al. (2023), Yan, Cheng, et al. (2023) developed a lightweight Cascading Residual Graph Convolutional Network that learns user preferences by continually refining embeddings across different types of behaviors. In addition, some scholars have attempted to apply deep learning methods to capture multi-type behavioral relations with temporal signals, thereby enhancing recommendation performance, such as DyMuS (Cho et al., 2023), PBAT (Su, Chen, et al., 2023), and TGT (Xia, Huang, Xu, Dai, & Bo, 2022).

Although deep learning-based multi-behavior recommendations have made significant progress, these models primarily focus on exploring the relevance between different behaviors using techniques such as Transformers and Graph Neural Networks, neglecting the influence of sparse interaction data within each behavior on model performance. For sparse data, existing methods either develop few-shot learning techniques (He, Chen, & Zhai, 2024; Su, Yan, et al., 2023) or perform data mining subsequent to augmentation (Han, Wu, et al., 2023). The former struggles to capture effective representations from limited data, while the latter inevitably introduces noise during the data expansion process, which may accumulate when multiple behaviors are aggregated. Hence, both approaches are unsuitable for multi-behavior recommendations. To address this, we propose a novel solution. Specifically, we employ implicit relationships to supplement sparse data and filter out noise, and on this basis, we aggregate node information within the simplex to explore strong correlations, *i.e.*, higher-order common features. Finally, by combining higher-order features across different behaviors, we achieve accurate multi-behavior recommendations.

2.2. Higher-order hypergraph neural network recommendation

Graph-based recommendation models construct input data in the form of graphs and design graph-learning models to make recommendations (Tao et al., 2020). Due to the inherent graph-structured nature of real-world interactions, graph neural network-based methods have been widely applied across various recommendation domains, such as point of interest recommendation (Meng et al., 2024; Qin et al., 2022; Yan, Song, et al., 2023), app recommendation (Hao et al., 2022; Ouyang, Guo, Wang, Liang, & Yu, 2022), click-through rate prediction (Li et al., 2021; Zhang, Wang, & Du, 2023), and so on Wang et al. (2023).

With the advancement of graph-based recommendation techniques, leveraging graph neural networks to express and explore higher-order interaction relationships has become an effective way to improve recommendation performance. For example, Wang, Ding, Hong, Liu, and Caverlee (2020) introduced a next-item recommendation framework, HyperRec, based on a hypergraph structure. This model utilizes a hypergraph to describe the relevance between items across multiple orders, and then mines and integrates multi-order connections within the hypergraph to achieve accurate recommendations. On the other hand, both Zhao et al. (2023) and Xia, Huang, Xu, Zhao, et al. (2022) combined hypergraphs with contrastive learning, effectively capturing deep-level cooperative relationships to boost recommendation performance. Yu et al. (2021) proposed a multi-channel hypergraph convolutional network that exploits higher-order relations to enhance social recommendations. Similar to Yu et al. (2021), He et al. (2021) employed HyperCTR to extract higher-order dependencies between items with multi-modal features. Beyond that,

Table 1
The descriptions of symbols employed in our study.

Symbols	Descriptions
U	set of users
S	set of items
B	set of behavior types
u	user $u \in U$
s	item $s \in S$
b	behavior $b \in B$
G_{pur}	purchasing graph $G_{\text{pur}} = \{V_{\text{pur}}, E_{\text{pur}}\}$
G_{bro}	browsing graph $G_{\text{bro}} = \{V_{\text{bro}}, E_{\text{bro}}\}$
G_{add}	adding-to-cart graph $G_{\text{add}} = \{V_{\text{add}}, E_{\text{add}}\}$
$\widetilde{u}_i$	the inter-behavior representation of user u_i
$\widetilde{s}_j$	the inter-behavior representation of item s_j
$\hat{u}_i$	the intra-behavior representation of user u_i
$\hat{s}_j$	the intra-behavior representation of item s_j
k	the number of iterations for implicit relationships
q	the order of simplices

hypergraph-based graph neural networks have also been widely applied to cross-domain recommendations (Wu, Long, Li, Yu, & Ng, 2022). For instance, Han, Zheng, Chen, Cheng, and Yao (2023) leveraged hypergraphs to model higher-order correlations between users and items across different domains, and proposed an intra- and inter-domain hypergraph convolutional network. This model identifies and mines higher-order relationships within domains, and jointly incorporates multi-domain features to alleviate data sparsity issues while enhancing recommendation performance. Unlike Han, Zheng, et al. (2023), Xu, Wei, et al. (2023) integrated multi-domain interactions into a unified graph, and presented a hierarchical hypergraph network-based correlative preference transfer framework for multi-domain recommendations (MDR). Although MDR and multi-behavior recommendations (MBR) differ in domain knowledge scope, their knowledge fusion and joint learning architectures are still worth exploring and studying.

Inspired by these studies, we aim to explore groups with higher-order relationships across different interactions, where members are closely interconnected and highly correlated. By identifying and aggregating higher-order relationships from diverse behaviors, we enhance the recommendation performance of the model. Specifically, we first mitigate data sparsity by learning implicit relationships within behaviors, and subsequently identify and aggregate features of nodes with higher-order relationships. It is worth noting that hypergraphs (Battiston et al., 2020) and simplicial complexes (Boccaletti et al., 2023; Xu, Feng, Zhu, & Xia, 2022) are two commonly used methods for studying higher-order relationships, frequently applied in fields such as evolutionary dynamics (Alvarez-Rodriguez et al., 2021; Majhi, Perc, & Ghosh, 2022; Xu, Wang, Xia, & Zhen, 2023) and disease propagation (Iacopini, Petri, Barrat, & Latora, 2019). Unlike hypergraphs, where each hyperedge can loosely contain multiple nodes, simplicial complexes have a more compact structure, with edges connecting each pair of nodes within a simplex, leading to stronger commonalities or similarities among nodes. Therefore, this study introduces simplicial complexes to enhance representation learning, which differs from previous research.

3. Research objectives

3.1. Notation definition

To provide a clear explanation of our model, we present the symbol descriptions used in HEM-GNN, as outlined in Table 1.

3.2. Problem definition

The objective of this study is to predict user preferences according to their various historical behaviors and recommend items aligned with their target behavior. Without loss of generality, we provide its mathematical description. For a given user set $U = \{u_1, u_2, \dots, u_{|U|}\}$, item set $S = \{s_1, s_2, \dots, s_{|S|}\}$, and behavior type set $B = \{b_1, b_2, \dots, b_t, \dots, b_{|B|}\}$, if there exists a type of interaction $b_n \in B$ between a user $u_i \in U$ and an item $s_j \in S$, we employ the triplet $\chi = (u_i, s_j, b_n)$ to record it. In particular, the behavior type set B contains $|B|$ different types of behaviors, where a specific type of behaviors is defined as the target behavior and the remaining $|B| - 1$ types are considered auxiliary behaviors. Our task is to generate a recommendation list $L_i = \{s_1, s_2, \dots\}$ for user u_i based on different types of historical interaction behavior, which matches his/her target behavior. The list is ranked based on the probability of user u_i interacting with different items under the target behavior, which is a classic regression problem. Therefore, it is crucial to precisely predict user preference under the target behavior, which directly determines the recommendation performance.

4. Methodology

In this subsection, we provide the overall architecture of HEM-GNN, as depicted in Fig. 2. This model primarily comprises four components: multi-behavior network construction, inter-behavior representation learning, intra-behavior representation learning and prediction layer.

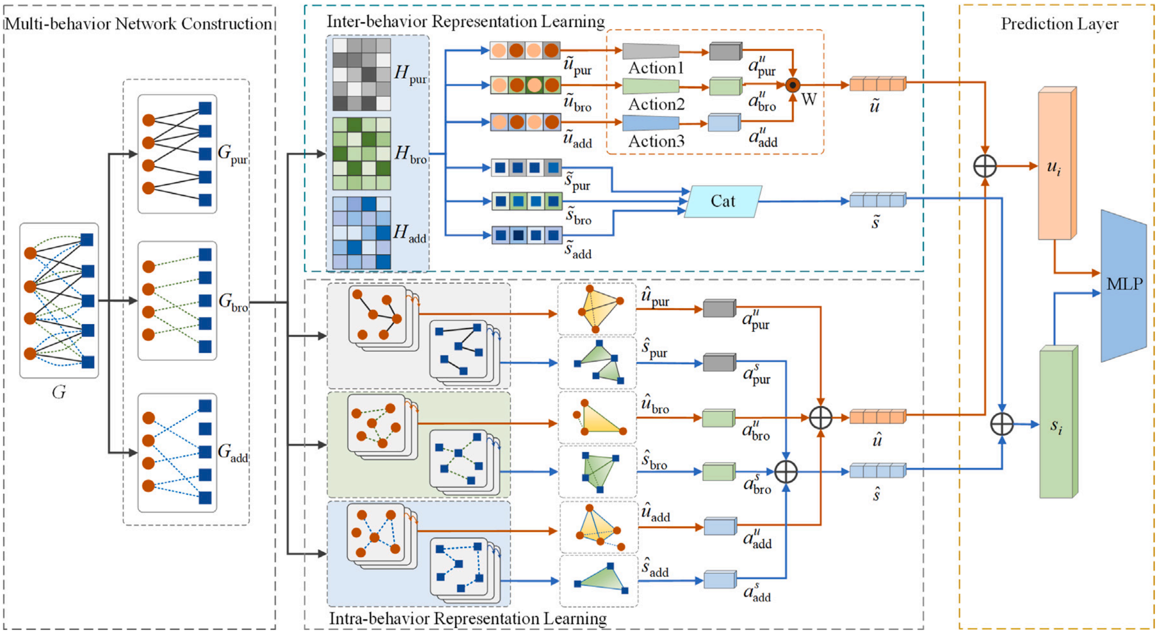


Fig. 2. Architecture of the HEM-GNN model.

- Multi-behavior network construction. In this part, we model the interactions between users and items as a weighted undirected network, in which nodes represent users or items and edges represent diverse types of interactions.
- Inter-behavior representation learning. To distinguish the significance of different behaviors and capture disparities among embeddings, HEM-GNN first extracts weight coefficients, and then integrates the embedding representations for various behaviors.
- Intra-behavior representation learning. For each behavior, HEM-GNN mitigates the issue of data sparsity by iteratively computing implicit relationships. Afterward, it leverages simplicial complexes to capture higher-order relationships among nodes and then perform information aggregation, thus enhancing the commonalities between nodes (individuals).
- Prediction layer. In this part, the features of individuals learned from inter- and intra-behaviors are fused, and then, HEM-GNN recommends items tailored to the target behavior.

4.1. Multi-behavior network construction

HEM-GNN constructs various types of interactions as distinct weighted bipartite graphs, including the purchasing graph G_{pur}, the browsing graph G_{bro}, the adding-to-cart graph G_{add}, and so forth. Taking the purchasing graph G_{pur} as an example, $G_{pur} = \{V_{pur}, E_{pur}\}$, where V_{pur} represents the set of nodes, $V_{pur} = \{U \cup S\}$, and E_{pur} denotes the collection of edges, with an edge $e_{ij} \in E_{pur}$ if a user u_i has purchased an item s_j.

Note that: (1) For the initial feature vectors of nodes, we utilize one-hot encoding to represent the interaction status of each node, where 1 and 0 respectively signify the presence or absence of interactions. One advantage of using one-hot encoding is its versatility in incorporating additional features of a user and an item (*e.g.*, user demographics and item attributes) (Gao et al., 2019). Meanwhile, to ensure consistency in the feature dimensions of users and items, we utilize embeddings to map the one-hot encoding into a 256-dimensional space. (2) For edge weights, we initialize the weights of edges in graphs (*e.g.*, browsing graph G_{bro} and purchasing graph G_{pur}) with the corresponding interaction frequency. For graphs without interaction counts, such as the bookmarking graph (where items are typically not bookmarked repeatedly), HEM-GNN defines a weight initialization function related to interaction time, as shown in Eq. (1).

$$w_{ij} = \frac{1}{\ln(D_{max} - D_{in} + 1) + 1}, \tag{1}$$

where D_{max} corresponds to the most recent date on which all interactions of the same type have occurred, and D_{in} denotes the date when the current interaction took place. The closer the interaction occurrence time, the larger the edge weight w_{ij}, and vice versa; $w_{ij} \in (0, 1]$.

4.2. Inter-behavior representation learning

For inter-behavior representation learning, HEM-GNN adopts the same architecture as the supervised task in S-MBRec (Gu, Wang, Shi, & Xiao, 2022) to perform supervised learning of weight coefficients for different behaviors. Specifically, HEM-GNN initially employs a spectral graph neural network to aggregate information among nodes under different behaviors. Subsequently, it learns the differences between various behaviors. Finally, it utilizes the acquired weights to fuse the features of users or items. The advantage of this approach is that it can effectively capture the disparities between behaviors while automatically learning weight coefficients to fuse the embeddings under each behavior with a supervised loss function to guide this task.

4.2.1. Node information aggregation

HEM-GNN separately learns the embedding representations of users and items under each behavior. Taking the purchasing graph G_{pur} as an example, we can obtain the adjacency matrix A_{pur} corresponding to the matrix R_{pur}, as illustrated in Eq. (2).

$$A_{\text{pur}} = \begin{pmatrix} 0 & R_{\text{pur}} \\ R_{\text{pur}}^{\top} & 0 \end{pmatrix}. \tag{2}$$

By leveraging the adjacency matrix A_{pur}, HEM-GNN conducts multi-layer information propagation and aggregation. The formula for information propagation is shown in Eq. (3).

$$H_{\text{pur}}^{(l+1)} = \sigma(\widetilde{D}^{-\frac{1}{2}}\widetilde{A}_{\text{pur}}\widetilde{D}^{-\frac{1}{2}}H_{\text{pur}}^{l}W_{\text{pur}}^{l}), \tag{3}$$

where $\widetilde{A}_{\text{pur}} = A_{\text{pur}} + I$, I indicates the identity matrix, $I \in \mathbb{R}^{|V|*|V|}$, $|V| = |U| + |S|$. $\widetilde{D}^{-\frac{1}{2}}\widetilde{A}_{\text{pur}}\widetilde{D}^{-\frac{1}{2}}$ is a normalized adjacency matrix with self-connections. $\widetilde{D}$ represents the degree matrix of $\widetilde{A}_{\text{pur}}$, $\widetilde{D} \in \mathbb{R}^{|V|*|V|}$. H_{pur}^{l} means the embedding matrix at the lth layer under the purchase behavior, $H_{\text{pur}}^{l} \in \mathbb{R}^{|V|*d}$. d denotes the embedding dimension. σ is a non-linear activation function. W_{pur}^{l} stands for the parameter of the model.

HEM-GNN aggregates information across all layers using the function F, which ensures that closer neighbors contribute more to the embeddings. The aggregation process is demonstrated in Eq. (4).

$$H_{\text{pur}} = F(H_{\text{pur}}^{l}), \tag{4}$$

where H_{pur} is composed of user embedding matrix $H_{\text{pur}}^{u} \in \mathbb{R}^{|U|*d}$ and item embedding matrix $H_{\text{pur}}^{s} \in \mathbb{R}^{|S|*d}$. F can be computed by weighted summation (He, Deng, Wang, Li, Zhang, & Wang, 2020) or cascading (Wang, He, Wang, Feng, & Chua, 2019). We evaluate the impact of these two methods on the recommendation performance of HEM-GNN, with the results presented in Appendix Figure S1. Based on the relevant literature (Gu et al., 2022) and our experimental results, we adopt the cascading approach in this study.

4.2.2. Inter-behavior difference learning

Different auxiliary behaviors have varying impacts on the target behavior. Hence, it is essential to combine the node embeddings under different auxiliary behaviors with the weight of the corresponding behavior. The weight learning should consider the scale and impact intensity of the corresponding behavior to distinguish between them. Taking the browsing graph G_{bro} as an example, the weight coefficient a_{bro}^{u} can be calculated by Eq. (5).

$$a_{\text{bro}}^{u} = \frac{exp(h_{\text{bro}} * n_{\text{bro}}^{u})}{\sum_{m\in B} exp(h_m * n_{\text{bro}}^{u})}, \tag{5}$$

where h_{bro} denotes the strength of browsing behavior, which is shared by all users. Here, we adopt the Hilbert–Schmidt Independence Criterion (HSIC) method (Gu et al., 2022; Huang & Wu, 2021) to calculate the similarity between auxiliary and target behaviors. Moreover, we investigate the performance of two alternative similarity measures - Cosine Similarity (COS) and Mutual Information (MI) - with the results presented in Appendix Figure S2. The experimental results demonstrate that HSIC is more suitable. n_{bro}^{u} indicates the number of times user u browses. It is evident that a_{bro}^{u} considers not only the strength of different behaviors for all users but also the proportion of a certain behavior of user u among all his/her behaviors. For the other auxiliary behaviors, we employ a similar approach.

4.2.3. Node feature fusion

Through the above procedures, we can acquire the embedding matrices of users/items for different behaviors along with their corresponding weights. HEM-GNN fuses the embeddings from all behaviors to generate the final representations of users and items. The representations aggregation of users from multiple behaviors is expressed by Eq. (6).

$$\widetilde{u}_i = \delta\Big\{W\Big(\sum_{m\in B} a_m^u * x_m^u\Big) + \varphi\Big\}, \tag{6}$$

where B represents the collection of behavior types; a_m^u denotes the weight coefficient; x_m^u indicates the representation learned by user u in behavior m; φ and W mean the biases and weights of the neural network, respectively; δ signifies the non-linear activation function.

Compared to users, the features of items tend to be more stable. Therefore, HEM-GNN adopts a concatenation operation to combine item representations from different behaviors, as described in Eq. (7).

$$\widetilde{s}_j = \mathcal{M}\Big\{Cat\Big(\sum_{m\in B} a_m^s * x_m^s\Big)\Big\}, \tag{7}$$

where B is the set of behavior types; a_m^s denotes the weight coefficient; x_m^s represents the representation of item s in behavior m; $Cat(\cdot)$ stands for the concatenation operation; $\mathcal{M}$ means a Multi-Layer Perceptron (MLP).

4.3. Intra-behavior representation learning

In each behavior, HEM-GNN should accurately identify similar neighbors of users/items and learn their corresponding features. Specifically, we start by converting the bipartite graphs of each behavior into homogeneous graphs and iteratively calculate implicit relationships among nodes. Next, we identify simplices within these homogeneous graphs and perform enhanced aggregation of node information. Finally, we merge node features from different behaviors to accomplish the representation learning for users/items.

4.3.1. Construction of homogeneous graphs and implicit relationships

HEM-GNN constructs distinct bipartite graphs based on behavior types. In this part, we transform each bipartite graph into two homogeneous graphs to explore relationships among nodes of the same type. For instance, we decompose the browsing graph G_{bro} into two homogeneous graphs: G^u_{bro} with nodes representing users and G^s_{bro} with nodes indicating items. In graph G^u_{bro}, if both user u_i and u_j browse the same item, there exists an edge e_{ij} with a weight greater than 0. The calculation of the initial weight w'_{ij} for edge e_{ij} is crucial, as it directly determines the exploration of relationships between nodes. HEM-GNN defines the calculation of the explicit relationship weight w'_{ij} between nodes i and j, as shown in Eq. (8).

$$w'_{ij} = \frac{y_{max} - \frac{1}{|T_{ij}|}\sum_{s_m \in T_{ij}} |y_{im} - y_{jm}|}{y_{max}}, \tag{8}$$

where y_{max} is the maximum weight of all edges in the browsing graph G_{bro}; T_{ij} represents the set of items in graph G_{bro} that are linked to both u_i and u_j, and $\left|T_{ij}\right|$ denotes the size of set T_{ij}; y_{im} indicates the edge weight between user u_i and item s_m in graph G_{bro}. w'_{ij} is calculated from the bipartite graph G_{bro} to determine the relationship between users u_i and u_j who have browsed the same item. Thus, w'_{ij} represents the explicit relationship. We use this value as the initial weight for the edge e_{ij} in the graph G^u_{bro}.

Additionally, in order to alleviate the issue of edge sparsity in the homogeneous graph G^u_{bro}, we use Eq. (9) to iteratively compute the implicit relationships among nodes.

$$\hat{w}_{ij} = \frac{\hat{y}_{max} - \frac{1}{|\hat{T}_{ij}|}\sum_{u_m \in \hat{T}_{ij}} |\hat{y}_{im} - \hat{y}_{jm}|}{\hat{Y}_{max}} * (1-\theta)^k, \tag{9}$$

where $\hat{y}_{max}$ indicates the maximum weight of all edges in the homogeneous graph G^u_{bro}; $\hat{T}_{ij}$ represents the collection of users in graph G^u_{bro} that are linked to both u_i and u_j; $\hat{y}_{im}$ means the edge weight between users u_i and u_m in graph G^u_{bro}; k signifies the number of iterative calculations; θ is the loss factor, which is a hyperparameter. Unlike w'_{ij}, $\hat{w}_{ij}$ is employed for further exploration of relationships between u_i and u_j in the homogeneous graph G^u_{bro}, and thus it is the implicit relationship.

After iteratively calculating $\hat{w}_{ij}$ according to Eq. (9), we utilize Eq. (10) to discard edges with relatively lower relevance (smaller weights).

$$\widetilde{w}_{ij} = \begin{cases} 0 & \widetilde{w}_{ij} \le \frac{\sum_{e_{ij} \in E^u_{\text{bro}}} |e_{ij}|}{|E^u_{\text{bro}}|}\xi \\ \widetilde{w}_{ij} & \text{others,} \end{cases} \tag{10}$$

where $\widetilde{w}_{ij}$ is the weight of the edge e_{ij} in the homogeneous graph G^u_{bro}; 0 indicates the edges to be discarded. E^u_{bro} and $\left|E^u_{\text{bro}}\right|$ represent the set and the number of edges in the homogeneous graph G^u_{bro}, respectively. ξ is a hyperparameter of the model. The calculation of edge weights in other homogeneous graphs also follows the aforementioned method.

4.3.2. Node information aggregation in simplices

In homogeneous graphs, we identify the highest-order simplex that each node belongs to and perform information aggregation on the nodes within each simplex. Definition of simplices: In the Euclidean space, given a set $A = \{r_i\}$ of geometrically independent points, if the set $\left[r_0, r_1, r_2, \dots, r_q\right]$ satisfies $\{r = \sum \lambda_i r_i | \lambda_i \ge 0, \sum \lambda_i = 1\}$, then $\left[r_0, r_1, r_2, \dots, r_q\right]$ is referred to as a q-dimensional simplex spanned by A, abbreviated as a q-simplex (Bianconi, 2021; Bick, Gross, Harrington, & Schaub, 2023). Simplicity, a 2-simplex is a triangle formed by three nodes, and a 3-simplex is a tetrahedron created by four vertices, and so on, as shown in Fig. 3.

Taking G^u_{bro} as an example, the highest-order simplex to which node u_i belongs refers to the largest set of nodes in G^u_{bro}, such that there is an edge between each pair of nodes, and this set includes u_i. The node information aggregation within a simplex is as depicted in Eq. (11).

$$u_i = f_{hi}\big(u_i, f_{lo}(\mathcal{N}^i_{sim})\big), \tag{11}$$

where u_i denotes the target node; $\mathcal{N}^i_{sim}$ represents the remaining nodes within the highest-order simplex to which node u_i belongs; f_{lo} is employed to aggregate information of $\mathcal{N}^i_{sim}$; f_{hi} is used to aggregate information between u_i and $f_{lo}(\mathcal{N}^i_{sim})$; both f_{lo} and f_{hi} are globally shared aggregation functions.

During the aggregation process of $f_{lo}(\mathcal{N}^i_{sim})$, we employ an attention mechanism to accurately capture the relationship between u_i and $u_j \in \mathcal{N}^i_{sim}$, as demonstrated in Eqs. (12)–(13).

$$f_{lo}\left(\mathcal{N}^i_{sim}\right) = \sum_{u_j \in \mathcal{N}^i_{sim}} u_j \times \frac{exp(Q_{ij})}{\sum_{u_m \in \mathcal{N}^i_{sim}} exp(Q_{im})}, \tag{12}$$

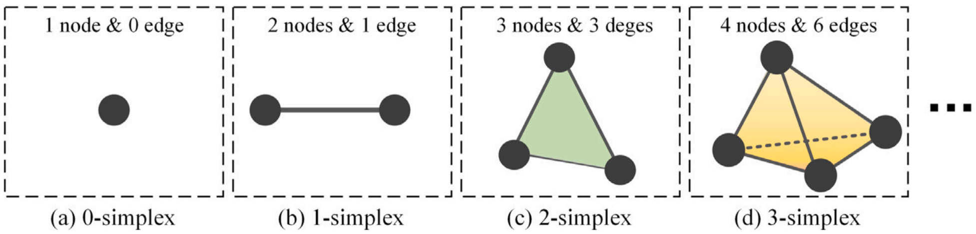


Fig. 3. Schematic illustration of a simplex structure. Within a simplex, there exists a relationship (link) between each pair of nodes. These connections foster a tighter simplex structure, indicating strong commonalities among the nodes. In general, we consider (a) 0-simplex and (b) 1-simplex as lower-order relations, while (c) 2-simplex, (d) 3-simplex, and so on are regarded as higher-order relations.

$$Q_{ij} = \left(u_i \odot u_j\right)^T tanh\left(w_{lo} \cdot Cat(u_i, u_j) + b_{lo}\right), \tag{13}$$

where $\odot$ represents Hadamard product; tanh is a nonlinear activation function. $Cat(\cdot)$ denotes the concatenation operation. w_{lo} and b_{lo} are hyperparameters. Q_{ij} reflects the significance of a neighbor $u_j \in \mathcal{N}^i_{sim}$ to the target node u_i.

HEM-GNN employs Eq. (14) to aggregate information between u_i and $f_{lo}(\mathcal{N}^i_{sim})$.

$$f_{hi}\left(u_i, f_{lo}\left(\mathcal{N}^i_{sim}\right)\right) = \phi\left(w_{hi} \cdot \left[u_i + f_{lo}\left(\mathcal{N}^i_{sim}\right) + b_{hi}\right]\right), \tag{14}$$

where ϕ is a nonlinear activation function; w_{hi} and b_{hi} are hyperparameters.

In addition, by stacking multiple aggregation layers, we can acquire more neighbor information and deeper representations, as illustrated in Eq. (15).

$$u^d_i = f_{hi}\left(u^{d-1}_i, (f_{lo}(\mathcal{N}^i_{sim}))^{d-1}\right). \tag{15}$$

4.3.3. Node feature fusion

Based on the aforementioned operations, HEM-GNN can learn user/item representations under different behaviors. For example, by leveraging graphs G^u_{bro} and G^u_{pur}, HEM-GNN can learn features of users in browsing and purchasing behaviors, respectively. In this part, HEM-GNN combines the features of nodes (users and items) across various behaviors. Taking users as an example, the process of feature fusion is demonstrated in Eq. (16).

$$\hat{u}_i = f_E\left(Cat(u^d_{i,\text{pur}}, \sum_{z \in B'} (a^u_{\text{pur}} \cdot u^d_{i,z}))\right), \tag{16}$$

where f_E represents the embedding operation; $u^d_{i,\text{pur}}$ denotes the dth-order representation of user u_i in graph G^u_{pur}; $Cat(\cdot)$ signifies the concatenation operation; B' means the collection of auxiliary behaviors; a^u_{pur} indicates the coefficient between the target and auxiliary behaviors, and its calculation is described in Eq. (5). Similarly, we can obtain the intra-behavior representation $\hat{s}_j$ for items.

4.4. Prediction layer

Through inter- and intra-behavior representation learning, we can obtain representations of users and items, denoted as $\widetilde{u}_i$, $\hat{u}_i$, $\widetilde{s}_j$, and $\hat{s}_j$. HEM-GNN utilizes these representations along with function Γ for predictions. Particularly, we adopt MLP with three hidden layers to implement the function Γ.

$$\hat{p}_{ij} = \Gamma(\widetilde{u}_i, \hat{u}_i, \widetilde{s}_j, \hat{s}_j). \tag{17}$$

4.5. Model learning

HEM-GNN recommends potential items to users for the target behavior by considering their multi-type historical behaviors, which is a regression problem. To better train the model, we employ the following objective function.

$$L = \frac{1}{|\mathcal{O}|} \sum_{(i \in U, j \in S)} (p_{ij} - \hat{p}_{ij})^2 + \lambda_l \|\Theta_l\|^2. \tag{18}$$

The loss function L comprises two components, where $\frac{1}{|\mathcal{O}|}\sum_{(i \in U, j \in S)}(p_{ij} - \hat{p}_{ij})^2$ evaluates the recommendation loss and $\lambda_l \|\Theta_l\|^2$ denotes the L2 regularization term to control the model's complexity and prevent overfitting. $\mathcal{O}$ represents the recommendation list. $|\mathcal{O}|$ means the length of $\mathcal{O}$. Θ_l indicates the collection of parameters in the framework.

4.6. Complexity analysis

The time complexity of HEM-GNN arises mainly from two parts: inter-behavior representation learning and intra-behavior representation learning.

(1) For inter-behavior representation learning, its time complexity mainly comes from the node information aggregation based on the spectral method. Taking the user graph G^u_{pur} for purchasing behavior as an example, the time complexity for eigendecomposition of the Laplacian matrix is $O((N^u_{\mathrm{pur}}+N^s_{\mathrm{pur}})^3)$, where N^u_{pur} and N^s_{pur} represent the number of users and items in G^u_{pur}, respectively. However, the eigendecomposition only needs to be performed once. As for the convolution operation, its time complexity is $O(h \cdot l \cdot (N^u_{\mathrm{pur}} + N^s_{\mathrm{pur}}) \cdot \tau)$, where h indicates the dimension of the embedding, l denotes the number of aggregation layers, and τ means the dimension of the hidden features.

(1) For intra-behavior representation learning, its time complexity primarily arises from the construction of the homogeneous graph and the node aggregation operation. Taking the user graph G^u_{pur} for purchasing behavior as an example, the construction of the homogeneous graph requires iteratively calculating the distance between nodes and their neighbors. Hence, its time complexity is $O(N^u_{\mathrm{pur}} \cdot M^u_{\mathrm{pur}} \cdot k \cdot h^2)$, where M^u_{pur} represents the average number of explicit neighbors for each user, and k is the number of iterations for computing implicit relationships. For node aggregation operation, its time complexity is $O(P^u_{\mathrm{pur}} \cdot q \cdot h^2)$, where P^u_{pur} signifies the number of q-simplices in G^u_{pur}, and q is the order of the simplex. Similarly, for intra-behavior representation learning in the item graph G^s_{pur}, the time complexity is $O(N^s_{\mathrm{pur}} \cdot M^s_{\mathrm{pur}} \cdot k \cdot h^2) + O(P^s_{\mathrm{pur}} \cdot q \cdot h^2)$, where N^s_{pur} represents the number of items in G^s_{pur}, M^s_{pur} means the average number of explicit neighbors, and P^s_{pur} stands for the number of q-simplices in G^s_{pur}.

It is worth noting that the values of l, q, and k are relatively small and can be ignored. For inter-behavior representation learning, we can adopt the simplified GCN proposed by Kipf and Welling (2017) to reduce its time complexity. Additionally, in both inter- and intra-behavior representation learning, different types of behaviors can be processed in parallel, which can significantly reduce the time complexity of the model. Finally, the construction of the graphs can be performed offline. Therefore, the time complexity of HEM-GNN is acceptable.

5. Experiments and analysis

We deploy the HEM-GNN model on the PyTorch platform, which has 8 NVIDIA A40 GPUs and 128 Hygon C86-7285 CPUs. In order to thoroughly evaluate and validate the performance of HEM-GNN, we conduct a series of experiments. In particular, we focus on the following research questions (RQs):

- RQ1: How does HEM-GNN's performance in multi-behavior recommendations compare to state-of-the-art algorithms?
- RQ2: How does HEM-GNN's performance under conditions of sparse data or cold start scenarios?
- RQ3: How does each component of HEM-GNN contribute to the enhancement of recommendation performance?

5.1. Experimental setup

5.1.1. Dataset

In this study, we validate our model using three publicly available datasets derived from the real world: Taobao, Beibei, and IJCAI. These datasets contain a variety of behavior types and interaction times. It is worth mentioning that these datasets are widely used in research related to multi-behavior recommendations (Chen et al., 2021; Ding, Yu, Li, He, & Jin, 2020; Xia, Huang, et al., 2021; Yang et al., 2022). We consider purchase behavior as the target behavior, while other behaviors are regarded as auxiliary behaviors. The statistics of these three datasets are presented in Table 2.

Taobao[1]: This dataset is sourced from Taobao, China's largest e-commerce platform. It consists of three types of interactions between users and products: browsing, adding-to-cart, and purchasing.

Beibei[2]: Similar to Taobao, this dataset also includes three types of interactions: browsing, adding-to-cart, and purchasing. The difference lies in that it is sourced from a popular e-commerce platform specializing in maternal and baby products, namely Beibei.com.

IJCAI[3]: This dataset, publicly released by the IJCAI competition, is utilized for the task of predicting repeat purchases. The dataset comprises four types of interactions: browsing, favoriting, adding-to-cart, and purchasing.

5.1.2. Comparison approaches

To validate the performance of HEM-GNN, we select two categories of algorithms to conduct comparative experiments: single- and multi-behavior recommendation models.

Single-behavior recommendation models:

NGCF (Wang et al., 2019) is a classical collaborative filtering recommendation model based on graph neural networks. This model constructs higher-order connectivity representations in user-item graphs and explicitly injects collaborative signals into user/item embeddings.

[1] https://tianchi.aliyun.com/dataset/dataDetail?dataId=649
[2] https://github.com/Scofield666/MBSSL/tree/master/dataset/Beibei
[3] https://tianchi.aliyun.com/dataset/42

Table 2
Statistics of the datasets.

Datasets	Taobao	Beibei	IJCAI
User	48,749	21,716	200,000
Item	39,493	7,977	808,354
Interactions	1,952,931	3,338,068	13,072,940
Number of interaction types	3	3	4
Target behavior	purchase	purchase	purchase
Auxiliary behaviors	browse,add-to-cart	browse,add-to-cart	browse, favorite,add-to-cart

ENMF (Chen, Zhang, Zhang, Liu, & Ma, 2020) is a recommendation method based on non-sampling strategies, and this model demonstrates strong robustness and generalization capabilities.

LightGCN (He et al., 2020) is an optimization and successor to NGCF. This model simplifies the design of graph neural networks, making it more suitable for recommendation tasks.

Multi-behavior recommendation models:

MATN (Xia et al., 2020) is an attention-based memory-augmented transformer network. This model achieves recommendation by jointly modeling collaboration signals for cross-type behaviors and contextual information for specific types of behaviors.

NMTR (Gao et al., 2019) integrates the Neural Collaborative Filtering (NCF) model with Multi-Task Learning (MTL) to extract user preferences from multi-behavior data.

EHCF (Chen, Zhang, Zhang, Ma, et al., 2020) is a state-of-the-art method dependent on non-sampling transfer learning. This model interconnects diverse behaviors in a transferable manner to capture complex relationships among them.

MB-GCN (Jin et al., 2020) is a typical and effective multi-behavior recommendation model based on graph convolutional networks. This model learns behavior strength through user-item propagation layers and captures behavior semantics, thus refining user/item embeddings.

MB-GMN (Xia, Xu, Huang, Dai, & Bo, 2021) is also a state-of-the-art multi-behavior recommendation model. It combines meta-learning and graph neural networks, utilizing a graph meta-network to capture multi-behavior signals and model dependencies among multiple behaviors.

S-MBRec (Gu et al., 2022) is a novel state-of-the-art recommendation model. This model learns the importance of different behaviors through supervised tasks and captures the commonalities of embeddings through contrastive learning tasks.

MBSSL (Xu, Wang, Wu, et al., 2023) employs a multi-behavior self-supervised learning framework to capture the diversity and interdependencies among multiple behaviors. It boasts state-of-the-art performance in the realm of multi-behavior recommendations.

5.1.3. Evaluation

(1) **Evaluation strategy.** For the three real-world datasets, we sort the interactions in ascending order based on their timestamps and then employ a leave-one-out strategy to conduct the performance evaluation. Specifically, we exclude each user's most recent purchase record and employ the remaining interaction records as the training set. Then, we randomly select 60% of users, considering their last purchase records as the test set, while the last purchase records of the non-selected users are designated as the validation set. In particular, similar to Chen, Zhang, Zhang, Ma, et al. (2020) and Xia, Xu, et al. (2021), we rank all items except positive ones for each user, which is more convincing than randomly sampling a subset of non-interactive items for each user (pairing each positive item instance with 99 randomly sampled items).

(2) **Evaluation metrics.** We employ the ranking metrics HR@K and NDCG@K, which are widely adopted in the multi-behavior recommendation field (Chen et al., 2021; Luo et al., 2023).

• HR@K is commonly used to measure the recall of top-K recommendations. It checks whether the target item is included in the recommendation list of length K. The calculation of HR@K is defined by Eq. (19).

$$\mathrm{HR}@K = \frac{Number\ Of\ Hits@K}{GT}, \tag{19}$$

where $Number\ Of\ Hits@K$ refers to the number of items in the recommendation list (of length K) that are of interest to the target user. GT denotes the total number of items of interest to the target user in the test set. HR@K is a specific form of recall, where a higher HR@K value indicates that the model has a higher probability of successfully predicting the target items within the top-K recommendations.

• NDCG@K is position-sensitive. In top-K recommendations, it is utilized to measure the ranking position of the target item in the recommendation list. The calculation of NDCG@K is shown as Eq. (20).

$$\mathrm{NDCG}@K = \frac{\mathrm{DCG}@K}{\mathrm{IDCG}@K}, \tag{20}$$

$$\mathrm{DCG}@K = \sum_{i=1}^{k} \frac{\mathrm{rel}_i}{\log_2(i+1)}, \tag{21}$$

$$\mathrm{IDCG}@K = \sum_{i=1}^{|\mathrm{REL}|} \frac{\mathrm{rel}_i}{\log_2(i+1)}, \tag{22}$$

Table 3
Hyperparameters settings of HEM-GNN.

Hyperparameters	Value	Description
h_{user}	256	user feature dimension
h_{item}	256	item feature dimension
θ	0.15	loss factor
ξ	0.5	weight coefficient
m	3	number of iterations for implicit relations
η	0.001	learning rate
λ_l	0.0001	balancing factor

Table 4
Performance comparison of our model with ten baseline methods on the Taobao dataset.

Models		HR@K				NDCG@K			
Types	Methods	K = 5	K = 10	K = 50	K = 100	K = 5	K = 10	K = 50	K = 100
Single	NGCF	.061(±.002)	.099(±.003)	.181(±.003)	.341(±.006)	.032(±.004)	.043(±.003)	.090(±.004)	.168(±.005)
	ENMF	.074(±.005)	.118(±.003)	.238(±.005)	.374(±.004)	.038(±.002)	.052(±.003)	.114(±.004)	.178(±.009)
	LightGCN	.093(±.004)	.123(±.005)	.261(±.005)	.410(±.010)	.070(±.003)	.066(±.004)	.126(±.003)	.193(±.006)
Multi	MATN	.114(±.004)	.137(±.003)	.290(±.004)	.446(±.009)	.081(±.003)	.079(±.003)	.144(±.005)	.213(±.004)
	NMTR	.125(±.006)	.174(±.008)	.302(±.005)	.547(±.007)	.082(±.009)	.097(±.004)	.151(±.004)	.248(±.008)
	EHCF	.148(±.007)	.214(±.006)	.373(±.012)	.613(±.014)	.091(±.004)	.116(±.004)	.189(±.005)	.278(±.004)
	MBCN	.185(±.003)	.250(±.003)	.451(±.008)	.711(±.012)	.103(±.003)	.143(±.003)	.209(±.003)	.327(±.013)
	MBGMN	.196(±.004)	.309(±.006)	.508(±.006)	.806(±.009)	.115(±.003)	.154(±.011)	.232(±.004)	.394(±.014)
	S-MBRec	.251(±.005)	.349(±.008)	.562(±.008)	.852(±.013)	.128(±.008)	.170(±.003)	.266(±.004)	.457(±.012)
	MBSSL	.283(±.006)	.364(±.005)	.578(±.007)	.877(±.011)	.141(±.004)	.181(±.004)	.280(±.003)	.471(±.009)
	HEM-GNN	*.320(±.006)	*.411(±.008)	*.626(±.007)	*.929(±.010)	*.162(±.004)	*.201(±.004)	*.301(±.004)	*.499(±.008)
	RI	13.12%	12.73%	8.23%	5.95%	14.45%	11.01%	7.42%	5.86%

'RI' means improvement percentage over the best-performing baseline.
* Indicates the significant improvement between our method and the baseline methods with $p < 0.05$.

where rel_i signifies the relevance score between the ith recommended item and the ground truth item, with $\mathrm{rel}_i \in [0, 1]$. REL denotes the optimal recommendation list, with |REL| representing its length. Unlike HR@K, NDCG@K not only focuses on whether the recommendation list contains items preferred by the user but also considers their positions in the list. That is, the higher the ground truth item rank, the greater the NDCG@K score.

In particular, following previous research on multi-behavior recommendations (Chen et al., 2021; Wang, Zhang, et al., 2020), we set the values of K to 5, 10, 50, and 100, respectively. Additionally, during the initial stage of recommendation, it is particularly important that the generated recommendation list contains items of interest to users. This helps to quickly build trust with users and enhance user retention rates. Therefore, in sparse data and cold start experiments, we employ the HR@K metric to evaluate the model's performance.

5.2. Experimental settings

For the models involved in our experiments, unless explicitly stated in the literature, we adopt the Adam optimizer and the Stochastic Gradient Descent (SGD) strategy for model training. We initialize the baseline algorithms with parameters obtained from the literature. Then, based on the dataset and optimization strategy, we fine-tune the models to achieve the optimal recommendation results. In particular, to ensure fairness of comparison, we set the embedding dimension of all models to 256. For the proposed HEM-GNN model, the order (q) of simplices is selected from {1, 2, 3, 4}, and the number of iterations (k) for implicit relationship computation is chosen from {1, 2, 3, 4, 5}. Finally, the parameters used in our model are listed in Table 3.

5.3. Empirical study

To compare HEM-GNN with ten baseline algorithms, we calculate the HR@K and NDCG@K metrics for each algorithm on three real-world datasets. Tables 4, 5, and 6 display the average recommendation accuracy and the corresponding standard deviations from experiments of 10 independent runs for all models. From the tables, we can infer the following conclusions.

(1) Our model outperforms other baseline algorithms on all three datasets, which demonstrates the effectiveness of HEM-GNN in multi-behavior recommendations. This is mainly attributed to: (a) In inter-behavior representation learning, HEM-GNN can capture the differences between auxiliary and target behaviors, which is beneficial for accurately predicting user preferences under the target behavior. (b) In intra-behavior representation learning, HEM-GNN introduces implicit relationships to alleviate data sparsity and filter noise, simultaneously facilitating the exploration of higher-order relationships among nodes under diverse behaviors. Additionally, by aggregating features of nodes in higher-order simplices, HEM-GNN enhances the commonalities among similar nodes. All of these operations are conducive to accurately learning user representations across different behaviors, thereby improving the precision of multi-behavior recommendations.

Table 5
Performance comparison of our model with ten baseline methods on the Beibei dataset.

Models		HR@K				NDCG@K			
Types	Methods	K = 5	K = 10	K = 50	K = 100	K = 5	K = 10	K = 50	K = 100
Single	NGCF	.058(±.004)	.085(±.005)	.174(±.003)	.366(±.009)	.031(±.003)	.042(±.004)	.086(±.005)	.190(±.003)
	ENMF	.066(±.003)	.096(±.005)	.227(±.004)	.549(±.013)	.033(±.003)	.044(±.005)	.119(±.004)	.248(±.006)
	LightGCN	.066(±.004)	.093(±.006)	.208(±.005)	.520(±.017)	.032(±.003)	.042(±.005)	.109(±.006)	.244(±.003)
Multi	MATN	.079(±.003)	.107(±.003)	.285(±.004)	.580(±.012)	.039(±.002)	.052(±.003)	.139(±.005)	.280(±.006)
	NMTR	.079(±.004)	.109(±.005)	.292(±.012)	.686(±.013)	.040(±.003)	.054(±.006)	.145(±.004)	.329(±.007)
	EHCF	.081(±.003)	.115(±.004)	.333(±.004)	.731(±.008)	.041(±.004)	.054(±.004)	.156(±.005)	.358(±.004)
	MBCN	.084(±.006)	.123(±.004)	.414(±.015)	.810(±.008)	.042(±.005)	.056(±.005)	.189(±.006)	.390(±.015)
	MBGMN	.091(±.005)	.140(±.004)	.444(±.003)	.860(±.010)	.043(±.005)	.067(±.005)	.200(±.005)	.424(±.011)
	S-MBRec	.093(±.004)	.161(±.005)	.459(±.011)	.884(±.014)	.044(±.002)	.080(±.005)	.223(±.006)	.430(±.014)
	MBSSL	.103(±.005)	.197(±.004)	.475(±.007)	.911(±.011)	.047(±.004)	.085(±.004)	.245(±.005)	.450(±.009)
	HEM-GNN	*.121(±.004)	*.231(±.004)	*.509(±.006)	*.921(±.009)	*.052(±.004)	*.093(±.005)	*.266(±.005)	*.469(±.007)
	RI	16.83%	17.16%	7.24%	1.07%	10.27%	9.41%	8.76%	4.22%

'RI' means improvement percentage over the best-performing baseline.
* indicates the significant improvement between our method and the baseline methods with $p < 0.05$.

Table 6
Performance comparison of our model with ten baseline methods on the IJCAI dataset.

Models		HR@K				NDCG@K			
Types	Methods	K = 5	K = 10	K = 50	K = 100	K = 5	K = 10	K = 50	K = 100
Single	NGCF	.003(±.0002)	.005(±.0003)	.011(±.0004)	.025(±.0007)	.001(±.0003)	.003(±.0005)	.007(±.0005)	.011(±.0004)
	ENMF	.006(±.0003)	.010(±.0003)	.018(±.0004)	.030(±.0014)	.004(±.0004)	.007(±.0006)	.011(±.0006)	.019(±.0005)
	LightGCN	.011(±.0003)	.014(±.0004)	.020(±.0014)	.034(±.0016))	.006(±.0004)	.010(±.0008)	.015(±.0005)	.022(±.0013)
Multi	MATN	.009(±.0004)	.012(±.0004)	.019(±.0006)	.033(±.0013)	.006(±.0005)	.009(±.0007)	.014(±.0008)	.021(±.0009)
	NMTR	.013(±.0009)	.016(±.0003))	.030(±.0005)	.055(±.0005)	.008(±.0004)	.011(±.0006)	.018(±.0007)	.030(±.0007)
	EHCF	.015(±.0005)	.021(±.0004)	.036(±.0005)	.058(±.0009)	.012(±.0004)	.015(±.0005)	.022(±.0006)	.033(±.0004)
	MBCN	.016(±.0004)	.022(±.0011)	.038(±.0004)	.060(±.0009)	.013(±.0005)	.015(±.0006)	.023(±.0005)	.035(±.0005)
	MBGMN	.018(±.0003)	.023(±.0006)	.039(±.0004)	.062(±.0012)	.014(±.0005)	.016(±.0006)	.024(±.0005)	.035(±.0007)
	S-MBRec	.020(±.0003)	.025(±.0003)	.042(±.0006)	.067(±.0012)	.015(±.0004)	.017(±.0007)	.027(±.0006)	.040(±.0004)
	MBSSL	.021(±.0004)	.027(±.0003)	.045(±.0005)	.068(±.0010)	.016(±.0004)	.018(±.0006)	.028(±.0004)	.042(±.0005)
	HEM-GNN	*.024(±.0003)	*.029(±.0004)	*.049(±.0006)	*.072(±.0009)	*.017(±.0006))	*.020(±.0004)	*.031(±.0005)	*.046(±.0005)
	RI	11.89%	10.07%	9.15%	4.81%	10.18%	7.68%	9.18%	9.08%

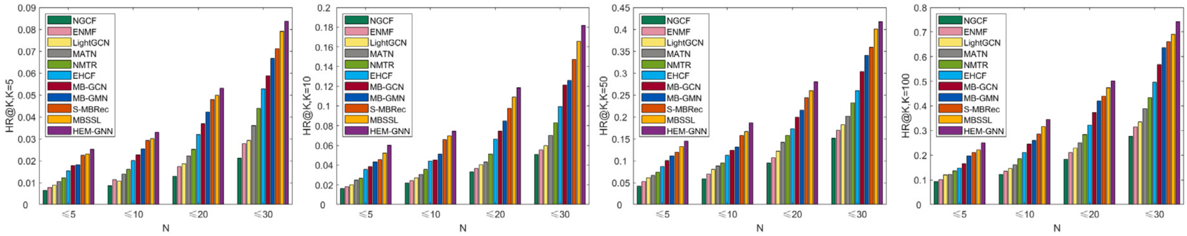
'RI' means improvement percentage over the best-performing baseline.
* indicates the significant improvement between our method and the baseline methods with $p < 0.05$.

(2) Overall, the performance of multi-behavior recommendation models is superior to that of single-behavior recommendation models. For example, in the Taobao and Beibei datasets, multi-behavior recommendation models such as MATN, NMTR, and EHCF perform better than single-behavior models like LightGCN. This suggests that distinguishing different types of user behaviors is meaningful. Interestingly, we find that the advantage of multi-behavior recommendation models over excellent single-behavior models does not always persist. For instance, in the IJCAI dataset, MATN performs slightly worse than LightGCN. We speculate this phenomenon stems from the fact that single-behavior recommendation models treat all auxiliary behaviors as the target behavior. Although this operation cannot distinguish the differences between different behaviors, introducing multiple behaviors can significantly alleviate the problem of data sparsity. For extremely sparse datasets, this may bring unexpected benefits. However, it is undeniable that this operation is still crude. Therefore, even so, the performance of LightGCN is still significantly worse than that of multi-behavior models like MBSSL.

(3) Among the baseline algorithms for multi-behavior recommendations, S-MBRec and MBSSL have the best performance. Both of them utilize a supervised learning framework to capture the relationships between multiple behaviors. Inspired by these algorithms, our model also adopts a similar architecture to extract differences among behaviors. The difference is that, within behaviors, we alleviate data sparsity by introducing implicit relationships and enhance higher-order features of individuals by aggregating information from nodes within simplices.

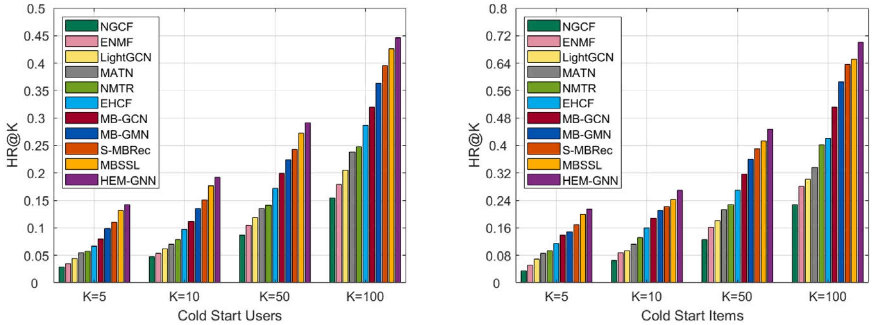
5.4. Data sparsity and cold start scenarios

Data sparsity and cold start pose the primary challenges for recommendation models. Thus, we conduct separate tests to evaluate the performance of HEM-GNN in these specific scenarios. In recommender systems, new users have limited attention and patience, typically focusing more on the top items in the recommendation list (Felfernig, Atas, Stettinger, Tran, & Leitner, 2024). This implies that in scenarios characterized by sparse data and cold starts, if the top few items can successfully hit user interests, it would greatly enhance user trust and satisfaction with the system. Therefore, in sparse data and cold start experiments, we leverage the hit rate HR@K to evaluate the performance of models.



(a) $K = 5$ (b) $K = 10$ (c) $K = 50$ (d) $K = 100$

Fig. 4. Recommendation performance of various models at 4 sparsity levels using Taobao dataset.



(a) (b)

Fig. 5. HR@K results under (a) user cold start and (b) item cold start using Taobao dataset.

5.4.1. Results in data sparse scenarios

We implement a random deletion strategy to spare the original data into four levels: approximately half of the users have less than 5, 10, 20, and 30 interactions for each behavior, respectively. We then conduct experiments using the sparsified data. Results for the Taobao dataset are displayed in Fig. 4, while those for the Beibei and IJCAI datasets are depicted in Appendix Figures S3 and S4, respectively.

For all models, as the sparsity of the dataset increases, their recommendation performance decreases to varying degrees. Taking the HEM-GNN model as an example, although it can utilize implicit relationships to enrich sparse data, the quality and quantity of these relationships are still influenced by the original data. It is worth mentioning that HEM-GNN outperforms all benchmark algorithms at different sparsity levels. There are two reasons for this advantage: (1) Using multiple behaviors, HEM-GNN can accurately discover the relationships between auxiliary and target behaviors. The incorporation of auxiliary behaviors partially alleviates the sparsity issue. (2) In intra-behavior representation learning, HEM-GNN enriches various types of interactions by computing implicit relationships, which greatly alleviates the data sparsity issue. Due to the above reasons, HEM-GNN exhibits outstanding performance on sparse data.

5.4.2. Results in cold start scenarios

Cold start is an extreme case of data sparsity, where personalized recommendations are offered to a target user without any data under the target behavior. It can be categorized into user cold start and item cold start. For user cold start, we choose specific users and remove their historical target behaviors, after which we provide recommendations to them. Similarly, for item cold start, we select certain items and remove their historical target behaviors, then recommend them to target users. Results for the Taobao dataset are shown in Fig. 5, while those for the Beibei and IJCAI datasets are presented in Appendix Figures S5 and S6, respectively.

In both user and item cold start scenarios, HEM-GNN outperforms baseline algorithms. This superior performance can be primarily attributed to HEM-GNN's ability to enrich auxiliary behaviors by leveraging implicit relationships, thereby accurately predicting user preferences for the target behavior. Interestingly, all tested models exhibit better performance in the item cold start scenario (Fig. 5(b)) than in the user cold start scenario (Fig. 5(a)). This is because items (products) have a certain level of stability. For instance, with lower-value products, users tend to browse and purchase immediately without much deliberation. In such cases, leveraging auxiliary behaviors like browsing can lead to better predictive outcomes. In contrast, user behavior patterns are more diverse – such as purchasing after browsing or buying after adding to cart – making it difficult for the model to make accurate predictions in the user cold start scenario.

5.5. Ablation experiment

To investigate the impact of different components within HEM-GNN on recommendation performance, we conduct ablation experiments. Specifically, we perform ablations on initialization weights, inter-behavior weight learning, intra-behavior implicit

Table 7
The settings of ablation models.

Ablation models	Description
HEM-In	The edge weights are initialized to 1 in interaction graphs for different types of behaviors. Other parameters are the same as HEM-GNN.
HEM-We	The weight coefficients a^u_{pur}, a^s_{pur}, a^u_{bro}, a^s_{bro}, a^u_{add}, a^s_{add} between different types of behaviors are all set to 1. Other parameters are the same as HEM-GNN.
HEM-Im	Implicit relationship extraction is omitted, *i.e.*, Eqs. (9)–(10) are not performed, and $k = 0$. Other parameters are the same as HEM-GNN.
HEM-Hi	Instead of the steps in Section 4.3.2, the GNN method (Scarselli, Gori, Tsoi, Hagenbuchner, & Monfardini, 2008) is employed to aggregate first-order neighbors of nodes. Other parameters are the same as HEM-GNN.

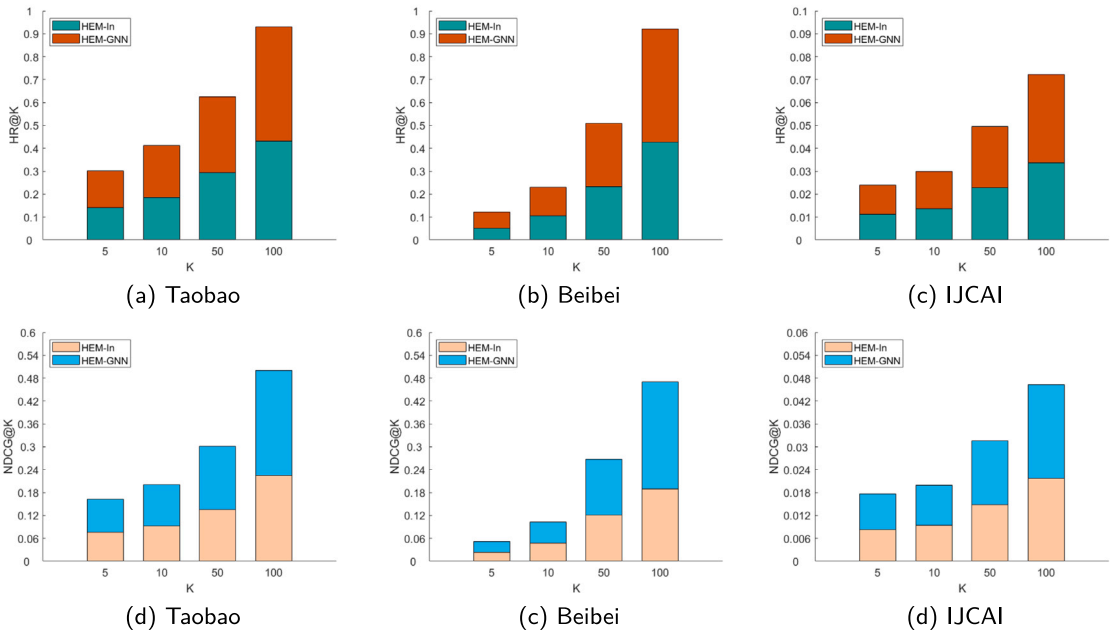


Fig. 6. Impact of initialization weight on the HEM-GNN model.

relationships, and higher-order enhancement, respectively, resulting in four new models (HEM-In, HEM-We, HEM-Im, and HEM-Hi) for comparison with HEM-GNN. The ablation settings of these four models are detailed in Table 7.

5.5.1. Initialization weight ablation

To validate the significance of the initial weights determined by HEM-GNN, we set all edge weights in the interaction graphs to 1, obtaining a new model called HEM-In. The recommendation performance of HEM-GNN and HEM-In is reported in Fig. 6.

It can be observed that HEM-In exhibits suboptimal performance compared to HEM-GNN. This can be attributed to the fact that both explicit and implicit relationship learning during intra-behavior representation learning are dependent on the initial edge weights. When all edge weights are equal, it becomes infeasible to identify important neighbors based on explicit relationships; and the supplementation of implicit relationships lacks directionality, introducing substantial noise. Consequently, the model struggles to accurately learn individual representations. In essence, the initial edge weights play a crucial role in exploring relationships within behaviors, which, in turn, impacts the extraction of higher-order features and the final recommendation accuracy.

5.5.2. Inter-behavior weight learning ablation

While keeping the model's input and output unchanged, we eliminate inter-behavior weight learning. In other words, we assign a uniform weight of 1 to all behavior types, resulting in a new recommendation model named HEM-We. Fig. 7 presents the recommendation performance of HEM-We and HEM-GNN.

As can be seen from Fig. 7, the performance of HEM-GNN is better than that of HEM-WE. This is because HEM-We directly fuses the representations of all types of behaviors using the same weights, without distinguishing between auxiliary behaviors and the target behavior. Although HEM-We learns representations of different behaviors separately, it treats all behaviors equally during representation aggregation. In other words, HEM-We crudely regards all behaviors as the target behavior, which will dilute truly valuable information and hinder the improvement of model performance. As a result, the performance of HEM-We is worse than that of HEM-GNN. This also highlights the significance of inter-behavior weight learning for the HEM-GNN model.

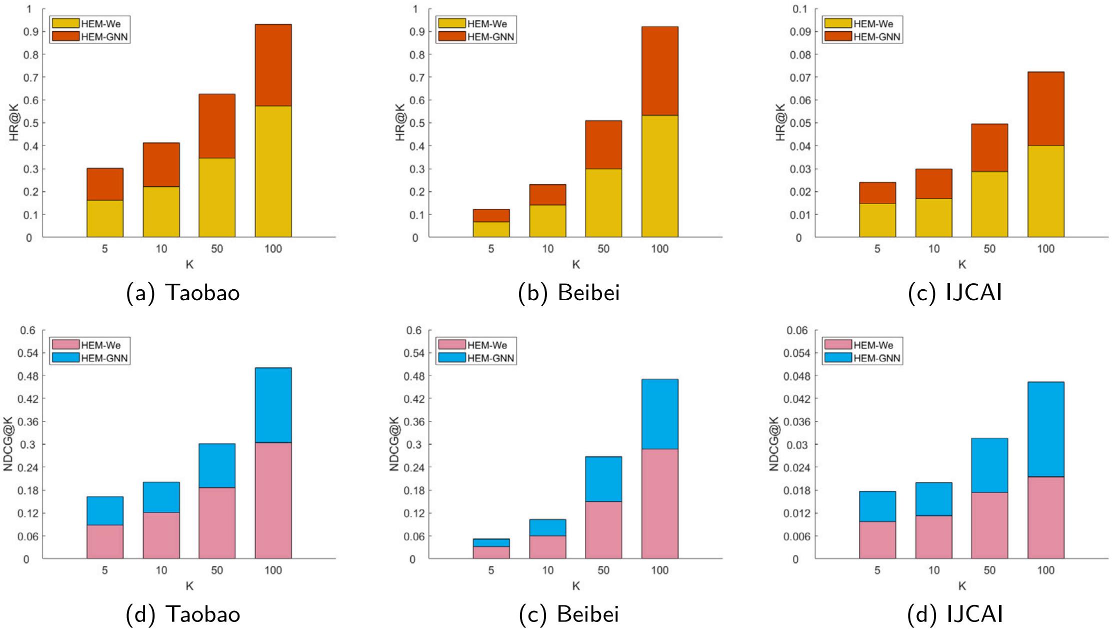


Fig. 7. Impact of inter-behavior weight learning on the HEM-GNN model.

5.5.3. Intra-behavior implicit relationships ablation

We remove the calculation of implicit relationships in HEM-GNN. That is, we directly utilize the original interaction data (explicit relationships) for intra-behavior representation learning, resulting in a new model called HEM-Im. The recommendation performance of HEM-Im and HEM-GNN is depicted in Fig. 8.

It is evident that HEM-GNN exhibits superior performance compared to HEM-IM. This is because HEM-GNN utilizes implicit relationships to enrich sparse data under different types of behaviors, which can effectively alleviate the issue of insufficient higher-order features. Additionally, the introduction of implicit relationships can filter out noise in explicit relationships. These factors lay the foundation for HEM-GNN to accurately learn node representations under different types of behaviors. In summary, implicit relationships, as a complement to explicit relationships, can significantly improve the recommendation performance of the model.

5.5.4. Higher-order enhancement ablation

In order to investigate the importance of aggregating information from nodes within a simplex in HEM-GNN, we replace it with a conventional GNN method. In this approach, we directly aggregate information from neighbors of target nodes to develop a novel method known as HEM-Hi. Fig. 9 depicts the recommendation performance of HEM-Hi and HEM-GNN.

As shown in Fig. 9, HEM-GNN outperforms HEM-Hi. This is because, unlike HEM-GNN, after enriching the data with implicit relationships, HEM-Hi directly aggregates all neighbor information of nodes using the graph neural network method. Excessive aggregation of node information may lead to over-smoothing and failure to capture subtle differences between nodes, which is ultimately detrimental to recommendation performance. In contrast, HEM-GNN leverages the simplicial complex technique to aggregate node information within simplices, thereby achieving an optimal balance between common and individual node representations, which contributes to improving recommendation performance. This indirectly verifies the importance of extracting higher-order common features under different types of behaviors for multi-behavior recommendation tasks.

5.6. Parameter sensitivity analysis

In this subsection, we conduct sensitivity experiments on important parameters of HEM-GNN, including the number of iterations (k) for implicit relationship calculation, the order (q) of simplices, and the number of layers in the MLP. Here, we hope that our model can optimize the ranking while ensuring the recall rate. Compared to HR@K, NDCG@K not only considers whether the target item is included in the recommendation list but also takes into account the ranking of the list. Therefore, we employ NDCG@K to evaluate the experimental results.

5.6.1. Iteration number of implicit relationships and order of simplices

In our model, as the number of iterations (k) for computing implicit relationships increases, new relationships are continuously generated to enrich sparse data, laying the foundation for extracting higher-order features. This suggests that the parameter k influences the order q of the extracted higher-order simplices. Therefore, we jointly evaluate these two parameters. Fig. 10 depicts

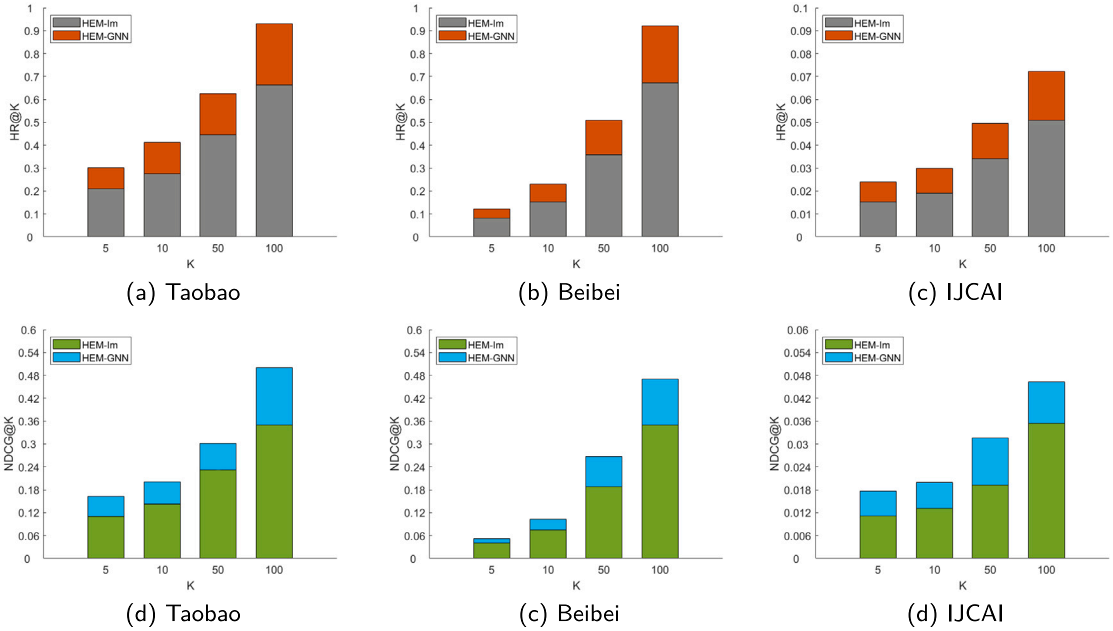


Fig. 8. Role of intra-behavior implicit relationships in the HEM-GNN model.

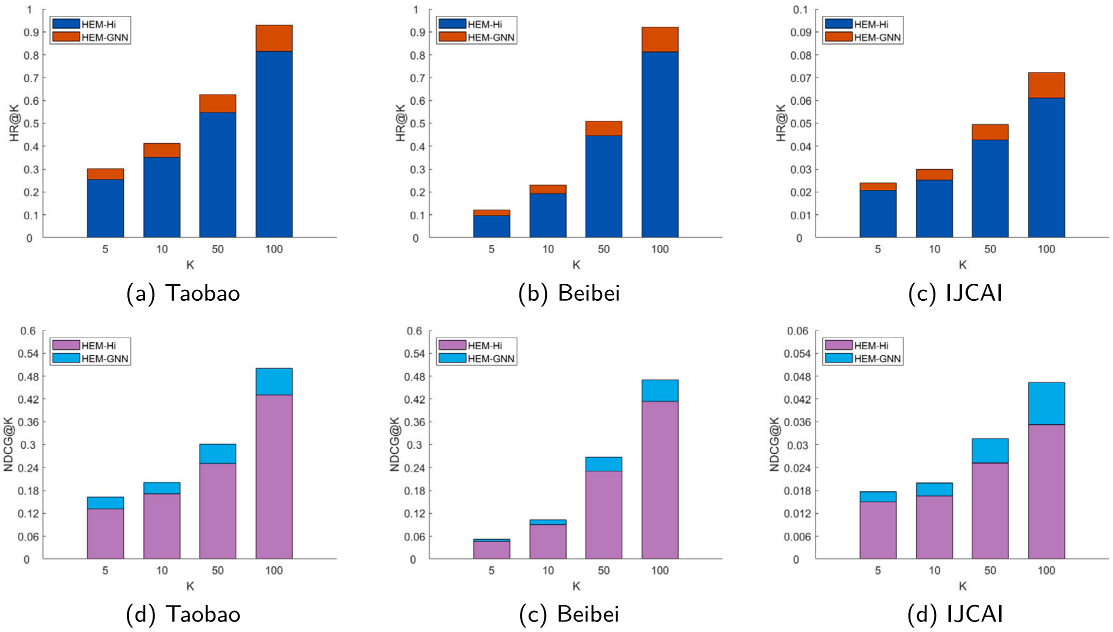
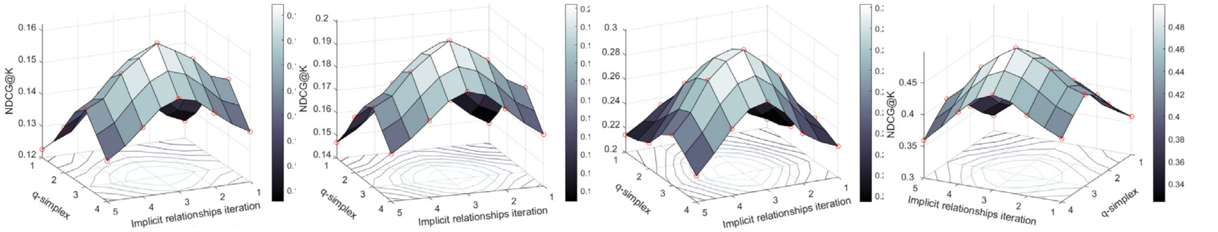


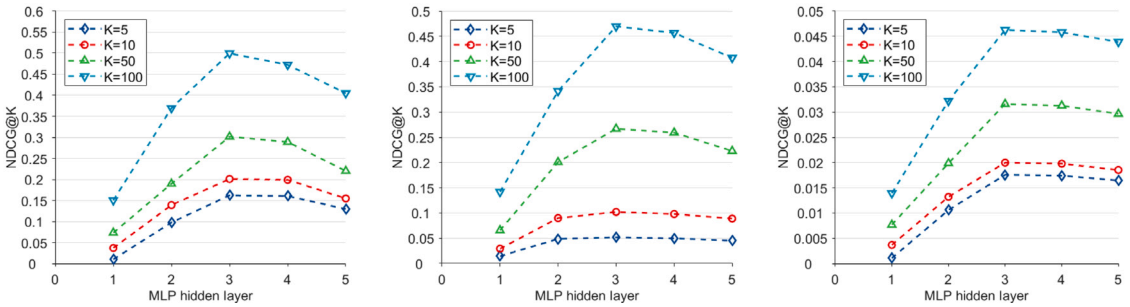
Fig. 9. Effect of higher-order enhancement on the HEM-GNN model.

the recommendation performance of HEM-GNN on the Taobao dataset across different values of k and q. It can be observed that HEM-GNN achieves optimal recommendation performance when $k = 3$ and $q = 3$. As k increases, the model's performance first rises and then declines. This is because, in the initial iterations, implicit relationships effectively alleviate data sparsity, thereby facilitating the extraction of higher-order features. However, with a continued increase in k, the noise contained within the implicit relationships gradually accumulates, which dilutes the effective features and leads to a decrease in model performance. Likewise, as q increases, the recommendation performance of HEM-GNN also first rises, then falls. This is because when q is smaller, the extracted higher-order features contain fewer nodes (*e.g.*, a 1-simplex includes two nodes, and a 2-simplex contains three nodes), resulting in more higher-order features, and thus the model becomes over-smoothing. Conversely, when q is larger, there are fewer



(a) $K = 5$ (b) $K = 10$ (c) $K = 50$ (d) $K = 100$

Fig. 10. Impact of implicit relationship iteration number and q-simplex on the recommendation performance of HEM-GNN (Taobao dataset). The red circles represent real experimental results, and the rest are fitting values.



(a) Taobao (b) Beibei (c) IJCAI

Fig. 11. Influence of the number of MLP layers on the recommendation performance of HEM-GNN.

high-order features, increasing the risk of overfitting. Similar conclusions can be drawn from the Beibei and IJCAI datasets (see Appendix Figures S7 and S8), with the exception that for IJCAI, HEM-GNN exhibits optimal performance at $k = 4$ and $q = 3$.

5.6.2. MLP layer

HEM-GNN employs the MLP for the final recommendation prediction. We conduct sensitivity experiments on the number of layers in the MLP, and the results are shown in Fig. 11. HEM-GNN achieves optimal performance with three hidden layers in the MLP. This is because as the number of layers in the MLP increases, the model gradually learns more complex and deeper features, enhancing its recommendation performance. However, with too many layers, the recommendation performance declines gradually. This can be attributed to overfitting, where the overly complex model becomes noise-sensitive, hampering the generalization ability of HEM-GNN.

6. Discussion and implications of our study

A series of experiments validate that HEM-GNN is an effective and reliable multi-behavior recommendation method. This achievement is primarily attributed to the model's proficiency in both inter- and intra-behavior representation learning. The former emphasizes exploring disparities among diverse behaviors, while the latter focuses on identifying unique individual traits within each behavior. It is worth mentioning that incorporating higher-order common features within each behavior can effectively enhance the performance of multi-behavior models. Moreover, our experimental results also validate that, for CF-based recommendations, improving the quality of representation is an effective strategy to enhance their overall performance.

From a theoretical perspective, we can conclude that: (1) In intra-behavior representation learning, the directly observed explicit relationships among individuals tend to be coarse-grained. Incorporating and utilizing implicit relationships can effectively supplement sparse data and filter out noise present in explicit relationships. There is no denying that the positive effects of explicit relationships take the main stage, but implicit relationships remain a valuable complement. (2) Compared to pairwise relationships, higher-order relationships are relatively fewer, yet their nodes exhibit stronger interconnections. Given the integration of diverse behavioral features in multi-behavior recommendations, emphasizing the quality of representation learning within each behavior becomes more crucial than focusing solely on quantity. (3) Exploring differences among behaviors serves as the fundamental basis for merging multiple behaviors, and it is also a key factor influencing recommendation performance.

From a practical perspective, we propose a general and scalable approach, named the simplices-based high-order enhanced graph neural network, for multi-behavior recommendations. This framework is also applicable across various recommendation domains, including but not limited to e-commerce, mobile applications, music recommendations, and more. Experimental results validate the superior performance of our model compared to 10 other baseline algorithms; this advantage persists in sparse data or cold start scenarios. In addition, our research offers valuable insights into the utilization of higher-order relationships in multi-behavior recommendation tasks. Also, the implicit relationship calculation and simplices-based higher-order aggregation method we proposed can also be extended to various fields, like social network analysis, among others.

7. Conclusions, limitations and outlook

In this study, we propose a simplices-based higher-order enhancement graph neural network for multi-behavior recommendations. This model extracts the differences and commonalities among individuals from both inter- and intra-behavior perspectives, respectively. For inter-behavior representation learning, HEM-GNN uncovers the distinctions between auxiliary and target behaviors. Meanwhile, for intra-behavior representation learning, it enriches and refines explicit relationships by introducing implicit ones. On this basis, HEM-GNN aggregates information from nodes within simplices to enhance higher-order commonalities. Finally, HEM-GNN utilizes the learned representations to complete accurate recommendations. Experimental results on three real-world datasets demonstrate the effectiveness of HEM-GNN. It is worth mentioning that, compared to baseline algorithms, HEM-GNN also has advantages in data sparse and cold start scenarios.

However, HEM-GNN also has certain limitations, primarily in three aspects: (1) In intra-behavior representation learning, HEM-GNN enriches and filters explicit relationships with implicit ones, followed by constructing homogeneous graphs for representation learning. However, these graphs are macroscopic and stable, making HEM-GNN lack attention to new and abrupt interactions. In other words, sudden interactions may be filtered out by implicit relationships, resulting in poor performance when predicting new, abrupt interactions. Certainly, as the new interaction frequency increases, HEM-GNN can gradually capture these features, but there may still be some delay. (2) When constructing relationship graphs, HEM-GNN pays more attention to frequency information, but lacks consideration for the sequence of interactions. It is well known that temporal information is a key factor influencing user behavior (Wang & Han, 2021). Therefore, in future work, we will incorporate temporal information into the HEM-GNN model to construct weighted graphs based on the sequence and frequency of interactions. The introduction of temporal information can effectively enhance the weight of recent interactions, thus alleviating the lag in HEM-GNN; meanwhile, it can also uncover the time-evolution features of user preferences, which is important for multi-behavior recommendations. (3) Undoubtedly, leveraging multiple user behaviors can expand the interaction data, from which we can extract high-quality higher-order features, which is helpful for us to accurately predict user preferences and make recommendations. At the same time, the introduction of multiple behaviors also increases the observation dimension of user behavior, not limited to the target behavior — purchasing, but also including auxiliary behaviors such as browsing, favoriting, etc. Whether these higher-order features implied in multiple behaviors can support us in providing reasonable explanations for recommended results from different perspectives. In other words, in future work, we can propose a graph neural network that can integrate higher-order features across multiple behaviors to offer valuable insights into the recommendation process, which is worth trying.

CRediT authorship contribution statement

Qingbo Hao: Writing – original draft, Visualization, Software, Methodology, Investigation. **Chundong Wang:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Yingyuan Xiao:** Writing – review & editing, Supervision, Conceptualization. **Hao Lin:** Writing – review & editing, Visualization, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the Joint Funds of the Tianjin Municipal Commission of Education, China (No. 2021YJSB252 and No. 2022BKY158); National Natural Science Foundation of China (No. U1536122); Tianjin Science and Technology Commission, China (No. 15ZXDSGX00030).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ipm.2024.103790.

References

Alvarez-Rodriguez, U., Battiston, F., de Arruda, G. F., Moreno, Y., Perc, M., & Latora, V. (2021). Evolutionary dynamics of higher-order interactions in social networks. *Nature Human Behaviour*, *5*(5), 586–595. http://dx.doi.org/10.1038/s41562-020-01024-1, URL https://www.nature.com/articles/s41562-020-01024-1.

Battiston, F., Cencetti, G., Iacopini, I., Latora, V., Lucas, M., Patania, A., et al. (2020). Networks beyond pairwise interactions: Structure and dynamics. *Physics Reports*, *874*, 1–92. http://dx.doi.org/10.1016/j.physrep.2020.05.004, URL https://www.sciencedirect.com/science/article/pii/S0370157320302489.

Bianconi, G. (2021). *Higher-order networks*. Cambridge University Press, http://dx.doi.org/10.1017/9781108770996, URL https://www.cambridge.org/core/elements/abs/higherorder-networks/26EE7ED4ECB114B2E231987C22759CDF.

Bick, C., Gross, E., Harrington, H. A., & Schaub, M. T. (2023). What are higher-order networks? *SIAM Review*, *65*(3), 686–731. http://dx.doi.org/10.1137/21M1414024, URL https://epubs.siam.org/doi/abs/10.1137/21M1414024.

Boccaletti, S., De Lellis, P., del Genio, C., Alfaro-Bittner, K., Criado, R., Jalan, S., et al. (2023). The structure and dynamics of networks with higher order interactions. *Physics Reports*, *1018*, 1–64. http://dx.doi.org/10.1016/j.physrep.2023.04.002, URL https://www.sciencedirect.com/science/article/pii/S0370157323001643.

Chen, C., Ma, W. Z., Zhang, M., Wang, Z. W., He, X. Q., Wang, C. Y., et al. (2021). Graph heterogeneous multi-relational recommendation. In *Proceedings of the 35th AAAI conference on artificial intelligence* (pp. 3958–3966). http://dx.doi.org/10.1609/aaai.v35i5.16515, URL https://ojs.aaai.org/index.php/AAAI/article/view/16515.

Chen, C., Zhang, M., Zhang, Y. F., Liu, Y. Q., & Ma, S. P. (2020). Efficient neural matrix factorization without sampling for recommendation. *ACM Transactions on Information Systems*, *38*(2), 1–28. http://dx.doi.org/10.1145/3373807, URL https://dl.acm.org/doi/10.1145/3373807.

Chen, C., Zhang, M., Zhang, Y. F., Ma, W. Z., Liu, Y. Q., & Ma, S. P. (2020). Efficient heterogeneous collaborative filtering without negative sampling for recommendation. In *Proceedings of the 34th AAAI conference on artificial intelligence* (pp. 19–26). http://dx.doi.org/10.1609/aaai.v34i01.5329, URL https://ojs.aaai.org/index.php/AAAI/article/view/5329.

Cheng, Z. Y., Han, S., Liu, F., Zhu, L., Gao, Z., & Peng, Y. X. (2023). Multi-behavior recommendation with cascading graph convolution networks. In *Proceedings of the ACM web conference* (pp. 1181–1189). http://dx.doi.org/10.1145/3543507.3583439, URL https://dl.acm.org/doi/abs/10.1145/3543507.3583439.

Cho, J. S., Hyun, D. M., won Lim, D., jae Cheon, H., Park, H.-i., & Yu, H. (2023). Dynamic multi-behavior sequence modeling for next item recommendation. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 4199–4207). http://dx.doi.org/10.1609/aaai.v37i4.25537, URL https://ojs.aaai.org/index.php/AAAI/article/view/25537.

Dai, W., Zhang, Q., Pan, W. K., & Ming, Z. (2019). Transfer to rank for top-n recommendation. *IEEE Transactions on Big Data*, *6*(4), 770–779. http://dx.doi.org/10.1109/TBDATA.2019.2892478, URL https://ieeexplore.ieee.org/abstract/document/8611201.

Ding, J. T., Yu, G. H., He, X. N., Feng, F. L., Li, Y., & Jin, D. P. (2021). Sampler design for bayesian personalized ranking by leveraging view data. *IEEE Transactions on Knowledge and Data Engineering*, *33*(2), 667–681. http://dx.doi.org/10.1109/TKDE.2019.2931327, URL https://ieeexplore.ieee.org/abstract/document/8778681.

Ding, J. T., Yu, G. H., He, X. N., Quan, Y. H., Li, Y., Chua, T.-S., et al. (2018). Improving implicit recommender systems with view data. In *Proceedings of the 27th international joint conference on artificial intelligence* (pp. 3343–3349). http://dx.doi.org/10.24963/ijcai.2018/464, URL https://www.ijcai.org/proceedings/2018/464.

Ding, J. T., Yu, G. H., Li, Y., He, X. N., & Jin, D. P. (2020). Improving implicit recommender systems with auxiliary data. *ACM Transactions on Information Systems*, *38*(1), 1–27. http://dx.doi.org/10.1145/3372338, URL https://dl.acm.org/doi/abs/10.1145/3372338.

Felfernig, A., Atas, M., Stettinger, M., Tran, T. N. T., & Leitner, G. (2024). Biases in group decisions. *Group Recommender Systems*, 153–163. http://dx.doi.org/10.1007/978-3-031-44943-7_8, URL https://link.springer.com/chapter/10.1007/978-3-031-44943-7_8#citeas.

Gao, C., He, X. N., Gan, D. H., Chen, X. N., Feng, F. L., Li, Y., et al. (2019). Neural multi-task recommendation from multi-behavior data. In *Proceedings of the IEEE 35th international conference on data engineering* (pp. 1554–1557). http://dx.doi.org/10.1109/ICDE.2019.00140, URL https://ieeexplore.ieee.org/abstract/document/8731537.

Gu, S. Y., Wang, X., Shi, C., & Xiao, D. (2022). Self-supervised graph neural networks for multi-behavior recommendation. In *Proceedings of the 31st international joint conference on artificial intelligence* (pp. 2052–2058). http://dx.doi.org/10.24963/ijcai.2022/285, URL https://www.ijcai.org/proceedings/2022/285.

Han, Y. Q., Wu, L. K., Wang, H., Wang, G. F., Zhang, M. D., Li, Z., et al. (2023). GUESR: A global unsupervised data-enhancement with bucket-cluster sampling for sequential recommendation. In *International conference on database systems for advanced applications* (pp. 286–296). http://dx.doi.org/10.1007/978-3-031-30672-3_19, URL https://link.springer.com/chapter/10.1007/978-3-031-30672-3_19.

Han, Z. X., Zheng, X. L., Chen, C. C., Cheng, W. J., & Yao, Y. (2023). Intra and inter domain hypergraph convolutional network for cross-domain recommendation. In *Proceedings of the ACM web conference* (pp. 449–459). http://dx.doi.org/10.1145/3543507.3583402, URL https://dl.acm.org/doi/abs/10.1145/3543507.3583402.

Hao, Q. B., Zhu, K., Wang, C. D., Wang, P., Mo, X. L., & Liu, Z. (2022). CFDIL: a context-aware feature deep interaction learning for app recommendation. *Soft Computing*, *26*(10), 4755–4770. http://dx.doi.org/10.1007/s00500-022-06925-z, URL https://link.springer.com/article/10.1007/s00500-022-06925-z.

He, L., Chen, H. X., Wang, D. X., Jameel, S., Yu, P., & Xu, G. D. (2021). Click-through rate prediction with multi-modal hypergraphs. In *Proceedings of the 30th ACM international conference on information & knowledge management* (pp. 690–699). http://dx.doi.org/10.1145/3459637.3482327, URL https://dl.acm.org/doi/abs/10.1145/3459637.3482327.

He, J. A., Chen, S. Q., & Zhai, J. D. (2024). POSTER: Pattern-aware sparse communication for scalable recommendation model training. In *Proceedings of the 29th ACM SIGPLAn annual symposium on principles and practice of parallel programming* (pp. 466–468). http://dx.doi.org/10.1145/3627535.3638481, URL https://dl.acm.org/doi/abs/10.1145/3627535.3638481.

He, X. N., Deng, K., Wang, X., Li, Y., Zhang, Y. D., & Wang, M. (2020). Lightgcn: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval* (pp. 639–648). http://dx.doi.org/10.1145/3397271.3401063, URL https://dl.acm.org/doi/abs/10.1145/3397271.3401063.

Huang, R., & Wu, Z. J. (2021). Multi-label feature selection via manifold regularization and dependence maximization. *Pattern Recognition*, *120*, Article 108149. http://dx.doi.org/10.1016/j.patcog.2021.108149, URL https://www.sciencedirect.com/science/article/pii/S0031320321003368.

Iacopini, I., Petri, G., Barrat, A., & Latora, V. (2019). Simplicial models of social contagion. *Nature Communications*, *10*(1), 2485. http://dx.doi.org/10.1038/s41467-019-10431-6, URL https://www.nature.com/articles/s41467-019-10431-6.

Jin, B. W., Gao, C., He, X. N., Jin, D. P., & Li, Y. (2020). Multi-behavior recommendation with graph convolutional networks. In *Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval* (pp. 659–668). http://dx.doi.org/10.1145/3397271.3401072, URL https://dl.acm.org/doi/abs/10.1145/3397271.3401072.

Kipf, T. N., & Welling, M. (2017). Semi-supervised classification with graph convolutional networks. In *Proceedings of the international conference on learning representations* (pp. 1–14). URL https://openreview.net/forum?id=SJU4ayYgl.

Li, Y. R., Guo, X. B., Lin, W. F., Zhong, M. J., Li, Q. W., Liu, Z. Y., et al. (2021). Learning dynamic user interest sequence in knowledge graphs for click-through rate prediction. *IEEE Transactions on Knowledge and Data Engineering*, *35*(1), 647–657. http://dx.doi.org/10.1109/TKDE.2021.3073717, URL https://ieeexplore.ieee.org/abstract/document/9409651.

Li, Q. F., Ma, H. F., Zhang, R. Y., Jin, W. Y., & Li, Z. X. (2023). Dual-scale contrastive learning for multi-behavior recommendation. *Applied Soft Computing*, Article 110523. http://dx.doi.org/10.1016/j.asoc.2023.110523, URL https://www.sciencedirect.com/science/article/pii/S1568494623005410.

Loni, B., Pagano, R., Larson, M., & Hanjalic, A. (2016). Bayesian personalized ranking with multi-channel user feedback. In *Proceedings of the 10th ACM conference on recommender systems* (pp. 361–364). http://dx.doi.org/10.1145/2959100.2959163, URL https://dl.acm.org/doi/abs/10.1145/2959100.2959163.

Luo, X., Wu, D. Q., Gu, Y. Y., Chen, C., Liu, L. C., Ma, J. W., et al. (2023). Criterion-based heterogeneous collaborative filtering for multi-behavior implicit recommendation. *ACM Transactions on Knowledge Discovery from Data*, *18*(1), 1–26. http://dx.doi.org/10.1145/3611310, URL https://dl.acm.org/doi/full/10.1145/3611310.

Majhi, S., Perc, M., & Ghosh, D. (2022). Dynamics on higher-order networks: A review. *Journal of the Royal Society Interface*, *19*(188), Article 20220043. http://dx.doi.org/10.1098/rsif.2022.0043, URL https://royalsocietypublishing.org/doi/full/10.1098/rsif.2022.0043.

Meng, L. Q., Liu, Z. Z., Chu, D. H., Sheng, Q. Z., Yu, J., & Song, X. Y. (2024). POI recommendation for occasional groups based on hybrid graph neural networks. *Expert Systems with Applications*, *237*, Article 121583. http://dx.doi.org/10.1016/j.eswa.2023.121583, URL https://www.sciencedirect.com/science/article/pii/S0957417423020857.

Meng, C., Zhang, H. Y., Guo, W., Guo, H. F., Liu, H. T., Zhang, Y. X., et al. (2023). Hierarchical projection enhanced multi-behavior recommendation. In *Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 4649–4660). http://dx.doi.org/10.1145/3580305.3599838, URL https://dl.acm.org/doi/abs/10.1145/3580305.3599838.

Ouyang, Y., Guo, B., Wang, Q. R., Liang, Y. J., & Yu, Z. W. (2022). Learning dynamic app usage graph for next mobile app recommendation. *IEEE Transactions on Mobile Computing*, *22*(8), 4742–4753. http://dx.doi.org/10.1109/TMC.2022.3161114, URL https://ieeexplore.ieee.org/abstract/document/9739893.

Qiao, S. T., Zhou, W., Luo, F. J., & Wen, J. H. (2023). Noise-reducing graph neural network with intent-target co-action for session-based recommendation. *Information Processing & Management*, *60*(6), Article 103517. http://dx.doi.org/10.1016/j.ipm.2023.103517, URL https://www.sciencedirect.com/science/article/pii/S0306457323002546.

Qin, Y. R., Gao, C., Wang, Y., Wei, S. Q., Jin, D. P., Yuan, J., et al. (2022). Disentangling geographical effect for point-of-interest recommendation. *IEEE Transactions on Knowledge and Data Engineering*, *35*(8), 7883–7897. http://dx.doi.org/10.1109/TKDE.2022.3221873, URL https://ieeexplore.ieee.org/abstract/document/9947308.

Qiu, H. H., Liu, Y., Guo, G. B., Sun, Z., Zhang, J., & Nguyen, H. T. (2018). BPRH: Bayesian personalized ranking for heterogeneous implicit feedback. *Information Sciences*, *453*, 80–98. http://dx.doi.org/10.1016/j.ins.2018.04.027, URL https://www.sciencedirect.com/science/article/pii/S0020025516315742.

Rendle, S., Freudenthaler, C., Gantner, Z., & Schmidt-Thieme, L. (2009). BPR: Bayesian personalized ranking from implicit feedback. In *Proceedings of the 25th conference on uncertainty in artificial intelligence* (pp. 452–461). URL https://dl.acm.org/doi/10.5555/1795114.1795167.

Scarselli, F., Gori, M., Tsoi, A. C., Hagenbuchner, M., & Monfardini, G. (2008). The graph neural network model. *IEEE Transactions on Neural Networks*, *20*(1), 61–80. http://dx.doi.org/10.1109/TNN.2008.2005605, URL https://ieeexplore.ieee.org/abstract/document/4700287.

Singh, A. P., & Gordon, G. J. (2008). Relational learning via collective matrix factorization. In *Proceedings of the 14th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 650–658). http://dx.doi.org/10.1145/1401890.1401969, URL https://dl.acm.org/doi/abs/10.1145/1401890.1401969.

Su, J. J., Chen, C. C., Lin, Z. B., Li, X., Liu, W. M., & Zheng, X. L. (2023). Personalized behavior-aware transformer for multi-behavior sequential recommendation. In *Proceedings of the 31st ACM international conference on multimedia* (pp. 6321–6331). http://dx.doi.org/10.1145/3581783.3611723, URL https://dl.acm.org/doi/abs/10.1145/3581783.3611723.

Su, L. C., Yan, F., Zhu, J. M., Xiao, X., Duan, H. Y., Zhao, Z., et al. (2023). Beyond two-tower matching: learning sparse retrievable cross-interactions for recommendation. In *Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval* (pp. 548–557). http://dx.doi.org/10.1145/3539618.3591643, URL https://dl.acm.org/doi/abs/10.1145/3539618.3591643.

Tao, Z. L., Wei, Y. W., Wang, X., He, X. N., Huang, X. L., & Chua, T.-S. (2020). Mgat: Multimodal graph attention network for recommendation. *Information Processing & Management*, *57*(5), Article 102277. http://dx.doi.org/10.1016/j.ipm.2020.102277, URL https://www.sciencedirect.com/science/article/pii/S0306457320300182.

Wang, J. L., Ding, K. Z., Hong, L. J., Liu, H., & Caverlee, J. (2020). Next-item recommendation with sequential hypergraphs. In *Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval* (pp. 1101–1110). http://dx.doi.org/10.1145/3397271.3401133, URL https://dl.acm.org/doi/abs/10.1145/3397271.3401133.

Wang, Y., & Han, L. X. (2021). Adaptive time series prediction and recommendation. *Information Processing & Management*, *58*(3), Article 102494. http://dx.doi.org/10.1016/j.ipm.2021.102494, URL https://www.sciencedirect.com/science/article/pii/S0306457321000054.

Wang, X., He, X. N., Wang, M., Feng, F., & Chua, T.-S. (2019). Neural graph collaborative filtering. In *Proceedings of the 42nd international ACM SIGIR conference on research and development in information retrieval* (pp. 165–174). http://dx.doi.org/10.1145/3331184.3331267, URL https://dl.acm.org/doi/abs/10.1145/3331184.3331267.

Wang, X., Liu, N., Han, H., & Shi, C. (2021). Self-supervised heterogeneous graph neural network with co-contrastive learning. In *Proceedings of the 27th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 1726–1736). http://dx.doi.org/10.1145/3447548.3467415, URL https://dl.acm.org/doi/abs/10.1145/3447548.3467415.

Wang, W., Zhang, W., Liu, S. K., Liu, Q., Zhang, B., Lin, L. Y., et al. (2020). Beyond clicks: Modeling multi-relational item graph for session-based target behavior prediction. In *Proceedings of the web conference* (pp. 3056–3062). http://dx.doi.org/10.1145/3366423.3380077, URL https://dl.acm.org/doi/abs/10.1145/3366423.3380077.

Wang, D., Zhang, X., Yin, Y., Yu, D., Xu, G., & Deng, S. (2023). Multi-view enhanced graph attention network for session-based music recommendation. *ACM Transactions on Information Systems*, *42*(16), 1–30. http://dx.doi.org/10.1145/3592853, URL https://dl.acm.org/doi/abs/10.1145/3592853.

Wu, H. R., Long, J. Y., Li, N. S., Yu, D. H., & Ng, M. K. (2022). Adversarial auto-encoder domain adaptation for cold-start recommendation with positive and negative hypergraphs. *ACM Transactions on Information Systems*, *41*(2), 1–25. http://dx.doi.org/10.1145/3544105, URL https://dl.acm.org/doi/abs/10.1145/3544105.

Xia, L. H., Huang, C., Xu, Y., Dai, P., & Bo, L. F. (2022). Multi-behavior graph neural networks for recommender system. *IEEE Transactions on Neural Networks and Learning Systems*, *Early Access*, 1–15. http://dx.doi.org/10.1109/TNNLS.2022.3204775, URL https://ieeexplore.ieee.org/abstract/document/9925059.

Xia, L. H., Huang, C., Xu, Y., Dai, P., Zhang, B., & Bo, L. F. (2020). Multiplex behavioral relation learning for recommendation via memory augmented transformer network. In *Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval* (pp. 2397–2406). http://dx.doi.org/10.1145/3397271.3401445, URL https://dl.acm.org/doi/abs/10.1145/3397271.3401445.

Xia, L. H., Huang, C., Xu, Y., Dai, P., Zhang, X. Y., Yang, H. S., et al. (2021). Knowledge-enhanced hierarchical graph transformer network for multi-behavior recommendation. In *Proceedings of the 35th AAAI conference on artificial intelligence* (pp. 4486–4493). http://dx.doi.org/10.1609/aaai.v35i5.16576, URL https://ojs.aaai.org/index.php/AAAI/article/view/16576.

Xia, L. H., Huang, C., Xu, Y., Zhao, J. S., Yin, D. W., & Huang, J. (2022). Hypergraph contrastive collaborative filtering. In *Proceedings of the 45th international ACM SIGIR conference on research and development in information retrieval* (pp. 70–79). http://dx.doi.org/10.1145/3477495.3532058, URL https://dl.acm.org/doi/abs/10.1145/3477495.3532058.

Xia, L. H., Xu, Y., Huang, C., Dai, P., & Bo, L. F. (2021). Graph meta network for multi-behavior recommendation. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval* (pp. 757–766). http://dx.doi.org/10.1145/3404835.3462972, URL https://dl.acm.org/doi/abs/10.1145/3404835.3462972.

Xin, X., Liu, X. Y., Wang, H. B., Ren, P. J., Chen, Z. M., Lei, J. H., et al. (2023). Improving implicit feedback-based recommendation through multi-behavior alignment. In *Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval* (pp. 932–941). http://dx.doi.org/10.1145/3539618.3591697, URL https://dl.acm.org/doi/abs/10.1145/3539618.3591697.

Xu, Y., Feng, M. L., Zhu, Y. Y., & Xia, C. Y. (2022). Multi-player snowdrift game on scale-free simplicial complexes. *Physica A. Statistical Mechanics and its Applications*, *604*, Article 127698. http://dx.doi.org/10.1016/j.physa.2022.127698, URL https://www.sciencedirect.com/science/article/pii/S0378437122004630.

Xu, J. C., Wang, C. K., Wu, C., Song, Y., Zheng, K., Wang, X. W., et al. (2023). Multi-behavior self-supervised learning for recommendation. In *Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval* (pp. 1–10). http://dx.doi.org/10.1145/3539618.3591734, URL https://dl.acm.org/doi/10.1145/3539618.3591734.

Xu, Y., Wang, J., Xia, C. Y., & Zhen, W. (2023). Higher-order temporal interactions promote the cooperation in the multiplayer snowdrift game. *Science China. Information Sciences*, *66*(222208), 1–16. http://dx.doi.org/10.1007/s11432-022-3738-3, URL http://scis.scichina.com/en/2023/222208.pdf.

Xu, Z. X., Wei, P. H., Liu, S. G., Zhang, W. M., Wang, L., & Zheng, B. (2023). Correlative preference transfer with hierarchical hypergraph network for multi-domain recommendation. In *Proceedings of the ACM web conference* (pp. 983–991). http://dx.doi.org/10.1145/3543507.3583331, URL https://dl.acm.org/doi/abs/10.1145/3543507.3583331.

Yan, M. S., Cheng, Z. Y., Gao, C., Sun, J., Liu, F., Sun, F. M., et al. (2023). Cascading residual graph convolutional network for multi-behavior recommendation. *ACM Transactions on Information Systems*, http://dx.doi.org/10.1145/3587693, URL https://dl.acm.org/doi/abs/10.1145/3587693.

Yan, X. D., Song, T. W., Jiao, Y. F., He, J. S., Wang, J. T., Li, R. P., et al. (2023). Spatio-temporal hypergraph learning for next POI recommendation. In *Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval* (pp. 403–412). http://dx.doi.org/10.1145/3539618.3591770, URL https://dl.acm.org/doi/abs/10.1145/3539618.3591770.

Yang, Y. H., Huang, C., Xia, L. H., Liang, Y. X., Yu, Y. W., & Li, C. L. (2022). Multi-behavior hypergraph-enhanced transformer for sequential recommendation. In *Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 2263–2274). http://dx.doi.org/10.1145/3534678.3539342, URL https://dl.acm.org/doi/10.1145/3534678.3539342.

Yu, J. L., Yin, H. Z., Li, J. D., Wang, Q. Y., Hung, N. Q. V., & Zhang, X. L. (2021). Self-supervised multi-channel hypergraph convolutional network for social recommendation. In *Proceedings of the web conference* (pp. 413–424). http://dx.doi.org/10.1145/3442381.3449844, URL https://dl.acm.org/doi/abs/10.1145/3442381.3449844.

Yu, J. L., Yin, H. Z., Xia, X., Chen, T., Li, J. D., & Huang, Z. (2024). Self-supervised learning for recommender systems: A survey. *IEEE Transactions on Knowledge and Data Engineering*, *36*(1), 335–355. http://dx.doi.org/10.1109/TKDE.2023.3282907, URL https://ieeexplore.ieee.org/abstract/document/10144391.

Zhang, W. F., Mao, J. W., Cao, Y., & Xu, C. F. (2020). Multiplex graph neural networks for multi-behavior recommendation. In *Proceedings of the 29th ACM international conference on information & knowledge management* (pp. 2313–2316). http://dx.doi.org/10.1145/3340531.3412119, URL https://dl.acm.org/doi/abs/10.1145/3340531.3412119.

Zhang, X., Wang, Z. M., & Du, B. (2023). Graph-aware collaborative reasoning for click-through rate prediction. *World Wide Web*, *26*(3), 967–987. http://dx.doi.org/10.1007/s11280-022-01050-1, URL https://link.springer.com/article/10.1007/s11280-022-01050-1.

Zhao, Z., Cheng, Z. Y., Hong, L. C., & Chi, E. H. (2015). Improving user topic interest profiles by behavior factorization. In *Proceedings of the 24th international conference on world wide web* (pp. 1406–1416). http://dx.doi.org/10.1145/2736277.2741656, URL https://dl.acm.org/doi/abs/10.1145/2736277.2741656.

Zhao, S., Wei, W., Mao, X. L., Zhu, S., Yang, M. H., Wen, Z. J., et al. (2023). Multi-view hypergraph contrastive policy learning for conversational recommendation. In *Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval* (pp. 654–664). http://dx.doi.org/10.1145/3539618.3591737, URL https://dl.acm.org/doi/abs/10.1145/3539618.3591737.